# UM-2: a 4+4 model of the Universe

Einstein-Lovelock gravity, the dirac16complex field, three generations of quarks and leptons with three neutrino flavours, their mixing and their flavour conversion, photons, gluons, and the $W^+$, $W^-$ and $Z$ bosons without a Higgs particle; every mass from the $x_0$ dependence of wave functions

Underlying theory: Patrick L. Nash  Derived and verified by Claude (Anthropic)

Model definition, version 2, 2026-10-18 (version 1: 2026-10-17)

Repository once-ere/Dirac_claude, folder universe_model_2

## Contents

**1 How to read this document** **2**

**2 Summary** **3**

**3 Space, coordinates and the periodic hidden direction** **4**
3.1 The coordinates (AUTHOR) . . . 4
3.2 The hidden coordinate $y$, the fold and the tips (DERIVED) . . . 4

**4 The metric as a power series** **5**
4.1 Order 0: the corrected primordial metric (AUTHOR coordinates, DERIVED properties) . . . 5
4.2 Order 1: the observable perturbations and the hidden tower . . . 5

**5 The action** **6**
5.1 Einstein-Lovelock gravity (AUTHOR weights) . . . 6
5.2 The block stabiliser (ASSUMED) . . . 6
5.3 The dirac16complex field . . . 6
5.3.1 The Clifford algebra of the author (AUTHOR, DERIVED verification) . . . 6
5.3.2 The Lagrangian and the canonical spin connection . . . 7
5.3.3 The two halves $\Psi_A$ and $\Psi_B$ and the fold parity (DERIVED) . . . 7
5.4 The Standard-Model gauge fields (ASSUMED structure, STANDARD content) . . . 7
5.5 The Standard-Model fermions (three generations; ASSUMED structure, INVENTED boundary assignment) . . . 8
5.6 The tip: electroweak symmetry breaking without a Higgs particle (INVENTED for this geometry) 9
5.7 The portal interactions of dirac16complex (INVENTED, gauge invariant) . . . 9
5.8 The catalogue of interactions . . . 10

**6 The $x_0$ mass mechanism** **12**
6.1 Gauge fields and the graviton (DERIVED) . . . 12
6.2 Fermions (DERIVED) . . . 12
6.3 The quanta of dirac16complex (DERIVED) . . . 15
6.4 What the uncertainty relation does and does not bound (DERIVED) . . . 15

**7 Dimensional reduction and the four-dimensional constants** **16**

**8 Gravity: the background, the stabilised block, the deflating extra times** **16**
8.1 The background field equations (DERIVED) . . . 16
8.2 The static block and attractive gravity with the author's weights (DERIVED) . . . 18

8.3 Stability: the radion (DERIVED) . . . 18
8.4 The deflating extra times (DERIVED, full nonlinear equations) . . . 18
8.5 The four-dimensional effective gravity (DERIVED) . . . 18

**9 dirac16complex in the cell: Kohn-Sham description 19**

**10 Three neutrino flavours: masses, mixing and flavour conversion (DERIVED) 20**
10.1 Masses and mixing . . . 20
10.2 The flavour evolution . . . 20
10.3 Vacuum, matter, the Sun and accelerators . . . 20
10.4 The early Universe and $N_{eff}$ . . . 21
10.5 The cosmic neutrino background . . . 22
10.6 Supernovae and the dense neutrino gas . . . 22
10.7 The $Z$ boson: invisible width and the widths of the three generations . . . 23

**11 The dark sector 23**

**12 Cosmology: CMB, CNB, BAO 24**

**13 Parameters 25**

**14 Assumptions, open problems and conflicts with data 26**

# 1 How to read this document

This document DEFINES the model UM-2. Every field, every interaction and every equation of the model is written out here; nothing has to be looked up elsewhere. Each statement carries one of these labels:

- **DERIVED**: computed and verified in this work (sympy or numerics; the verification is stated with the result);
- **AUTHOR**: taken from the author's notebooks (coordinates, metric, Lovelock weights, gamma matrices, mass parameter);
- **STANDARD**: textbook physics used as an input;
- **ASSUMED**: a choice made for this model;
- **INVENTED**: a new phenomenological ingredient of this model;
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

"(record)" after ASSUMED marks a choice taken over from the Revision record of this repository. Units: $\hbar = c = 1$ unless stated. Indices $M, N, \ldots$ run over the eight coordinates, $a, b, \ldots$ over the orthonormal frame, $\mu, \nu$ over the four observable coordinates $x_1, x_2, x_3, x_4$.

**Version 2 (2026-10-18).** Version 1 contained one Standard-Model generation and used a band of dirac16complex as a stand-in for the missing neutrinos $\nu_\mu$, $\nu_\tau$. Version 2 contains three full generations of quarks and leptons, anomaly free generation by generation, with three neutrino flavours, their masses, their mixing (the PMNS matrix) and their flavour conversion in vacuum, in matter, in the Sun, in the early Universe and in supernovae; the quark mixing (the CKM matrix) follows from the same mechanism. The chiral tip $Z$ portal of version 1 is removed (section 5.7). Every other part of the model is unchanged.

Symbols with more than one use:

- $k$ is the block variable $1/f^2$ ($k_*$, $k_\pm$, $\delta k$, $G(k)$); as the index of $\alpha_k$, $E_{(k)}$ and $\mathcal{L}_{(k)}$ it is the Lovelock order (written $p$ in the reduction formula of section 8.5, where the block variable appears as well); in $e^{ikx_1}$, in $h_j$, in $\vec{k}$, in $\Delta k$ and in the dispersion $\varepsilon = e^{-a_4}|k|$ it is the momentum of a mode along $x_1, x_2, x_3$.
- $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and $\Lambda_4$ the four-dimensional one; in the $x_0$ eigenvalue problems $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue ($\Lambda_n$, $\Lambda_W$, $\Lambda_Z$, $\Lambda_D$ are particular ones); $\Lambda_\star$ is the scale of the effective interactions; $\Lambda_\ell$ and $\Lambda_q$ are the tip coupling matrices.
- $n$ is a number density ($n_o$ that of a Kohn-Sham orbital), except as a level or period number ($\Lambda_n$, $z \to n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell.

- The subscript $t$ ($x_t$, $p_t$, $v_t$) refers to the extra times $x_5, x_6, x_7$; $\xi = y/L$ is the scaled hidden coordinate.
- $f$ is the block scale and $f_o$ an occupation number. $\rho$ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$; $\varrho$ is a probability density in $\xi$.
- $w = (w_1, w_2, w_3)$ are the Lovelock weights; $w$ is also the weight 1/2 of a fold parity in the Kohn-Sham sums, $\sqrt{c^2 - e^2}$ in the fermion mass condition, and the equation-of-state parameter $p_3/\rho$. $z = 6Hx_0$, except where it is a redshift (dark energy, cosmology).
- Flavour: $U$ is the PMNS matrix and $V$ the CKM matrix (the gauge transformations are $U_L$, $U_Y$, the block basis $V_\beta$, the transverse volume $V_\perp$, and the matter potentials always carry a subscript, $V_{CC}$, $V_{NC}$); $\kappa_\nu$, $\kappa_e$, $\kappa_u$, $\kappa_d$ are the $3 \times 3$ tip matrices of the four fermion sectors ($\kappa_f$ the tip mixing of one generation, $\kappa = 8\pi G_8$ without subscript); $\theta_{12}$, $\theta_{13}$, $\theta_{23}$, $\delta$ are mixing angles and the Dirac phase ($\theta_W$ the weak angle, $\theta$ also the parameter of a gauge transformation; $\delta$ also a variation, as in $\delta k$), $s_{ij} = \sin\theta_{ij}$ and $c_{ij} = \cos\theta_{ij}$; $J$ is the Jarlskog invariant, except in the block operator $J = \gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$.
- Generations and flavours: $i = 1, 2, 3$ is the generation index of the fermion fields and labels the neutrino mass states (in the curvature formulas $i, j$ are 3-space indices); $\alpha \in \{e, \mu, \tau\}$ is a flavour index (on $G^\alpha$ and $T^\alpha$ a colour index).
- Fermion names: $e$, $\mu$, $\tau$ are the charged leptons, $\nu_1, \nu_2, \nu_3$ the neutrino mass states and $\nu_e, \nu_\mu, \nu_\tau$ the flavour states; $u, c, t$ and $d, s, b$ the quarks. The same letters have other meanings: $e$ is also the elementary charge and the dimensionless fermion eigenvalue $\varepsilon fL$; $\mu_\psi$ is a bulk mass and $\mu = \sqrt{2}G_F n_{\nu_e}$ the unit of the neutrino self-interaction; $c$ with a subscript ($c_L$, $c_Q$, $c_R$, $c_\psi$) is a localisation $f\mu L$; $s$ is also $\sinh w/w$ in the fermion mass condition; $t$ as a subscript refers to the extra times.
- $H$ is the author's constant (an inverse length), except in the neutrino-decoupling equations, where $H = \sqrt{8\pi\rho/3}/M_{Pl}$ is the Hubble rate, $M_{Pl} = G_N^{-1/2} = 1.22091 \times 10^{19}$ GeV the Planck mass and $\rho$ the total energy density. The time is always $x_4$, and the symmetry axis of the fast-flavour analysis is $x_3$. $\Lambda^0$, $\Lambda^3$ (upper indices) are the potentials of the fast-flavour analysis, $\Lambda_0 = -24H^2$ the cosmological constant of the sourceless block. $G(k)$ is the Einstein-Hilbert coefficient and $G(u)$ the angular distribution of the electron lepton number; $F_h = 1.0095$ is the heating factor of $\Delta N_{eff}$ and $F(T_1, T_2)$ the energy-transfer kernel of neutrino decoupling.

# 2 Summary

1. **Spacetime.** Eight dimensions with the author's coordinates $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$: $x_0$ is the hidden space-like direction, $x_1, x_2, x_3$ are ordinary space, $x_4$ is the time, and $x_5, x_6, x_7$ are three extra time-like directions. The frame metric is $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in this order. The metric depends on $x_0$ only through $z = 6Hx_0$ with period $\pi$ in $z$, i.e. period $\ell_0 = \pi/(6H)$ in $x_0$.
2. **Gravity.** Einstein-Lovelock gravity in eight dimensions with the author's weights $w = (1, 1/32, 1/1152)$ and $\Lambda = -10H^2$, kept unchanged. A block stabiliser (a vacuum energy of the hidden block, ASSUMED) gives a stable static block at $k_* = 12 + \frac{4}{3}\sqrt{51}$ on which four-dimensional gravity is attractive: its Einstein-Hilbert coefficient is $G(k_*) = \frac{20}{3} + \sqrt{51} = 13.81 > 0$ (DERIVED).
3. **The deflating extra times.** In the primordial era the extra times $x_5, x_6, x_7$ (and $x_0$ with them) deflate at rates up to $2.6H$ into the stabilised block, then the block oscillates with the radion frequency $3.387H$ while 3-space keeps expanding (DERIVED, full nonlinear equations).
4. **dirac16complex** $\Psi$: a 16-component complex anticommuting spinor of $Cl(4,4)$ (Pin(4,4) irreducible; two inequivalent 8-dimensional irreducible representations of Spin(4,4)) coupled to gravity through the canonical spin connection, which is non-trivial: $\gamma^M \Omega_M = \frac{3}{2}(H/f)\gamma^{x_0} + (\dot{b}_4/b_4 - \frac{1}{2}\dot{a}_4)\gamma^{x_4}$. It splits into two Lorentz-invariant halves, $\Psi_A$ ($B = -1$) and $\Psi_B$ ($B = +1$), with $B = -iC\gamma^{x_4}$ the Krein metric.
5. **Standard-Model content (three generations, no Higgs particle).** Photon, gluons, $W^+$ (charge $+e$), $W^-$ (charge $-e$), $Z$ (neutral), and three generations of quarks, $(u, d)$, $(c, s)$, $(t, b)$, and of leptons, $(\nu_e, e)$, $(\nu_\mu, \mu)$, $(\nu_\tau, \tau)$, each generation with a right-handed neutrino. The gauge anomalies cancel generation by generation. Electroweak symmetry is broken at the regular tip of the hidden direction by a non-linear sigma field $\Sigma \in SU(2)$ that has no radial mode, hence no Higgs particle.
6. **Masses and mixing.** Every four-dimensional mass is the eigenvalue of an $x_0$ operator acting on a wave function periodic in $x_0$ with period $\pi/(6H)$. Massless: graviton, photon, gluons (flat $x_0$ profiles) and the fold band of $\Psi$ (chiral zero modes). Massive: $W^\pm$, $Z$ (profiles bent at the tip), the twelve fermions (a doublet zero mode and a singlet zero mode of each fermion mixed at the tip by a $3 \times 3$ tip matrix in each sector), the bulk band of $\Psi$ and every Kaluza-Klein mode (DERIVED spectra, fitted to the measured masses with order-one parameters). The doublet bulk mass is common to the three generations of a sector; then the left-handed

mixing matrices are $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$ (DERIVED, exact); with $\kappa_e = \kappa_u = 1$ (ASSUMED) they are the tip matrices $\kappa_\nu$ and $\kappa_d$, whose values are the measured PMNS and CKM matrices (STANDARD inputs).

7. **Three neutrino flavours.** Dirac neutrinos with masses 1.000, 8.712 and 50.140 meV (normal ordering, the NuFIT 6.0 mass splittings, the lightest mass ASSUMED) and the PMNS mixing of NuFIT 6.0 (Dirac phase 212°). Their flavour conversion is computed in vacuum, in matter (the MSW effect), in the Sun, at accelerators, in the early Universe (where it accompanies neutrino decoupling, $N_{eff} = 3.0432$), in the cosmic neutrino background (three mass eigenstates) and in supernovae (the MSW resonances of the envelope, the fast flavour instability of the dense neutrino gas and its effect on the explosions).
8. **New interactions (INVENTED).** A gauge-kinetic portal couples $\Psi$ to pairs of gauge bosons ($W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$, gluon pairs); a bounded vector contact portal couples $\Psi$ to the fermion currents. The chiral tip $Z$ portal of version 1 is removed: with three neutrinos it has no role, it is bounded by the $Z$ width and disfavoured by $N_{eff}$ (section 5.7).
9. **The dark sector.** The lightest odd states of the two halves $\Psi_A$ and $\Psi_B$, degenerate Dirac fermions of mass $3.1022\,m = 44.35$ TeV ($m = H$), are electrically and weakly neutral, stable by the fold parity, cold, and together a dark-matter candidate; their relic abundance, however, cannot be produced by the gauge-kinetic portal within perturbative unitarity (section 11; OPEN). No state of $\Psi$ is dark energy; the dark energy of UM-2 is a tuned four-dimensional cosmological term.
10. **Cosmology.** The four-dimensional effective theory of the stabilised block is exact general relativity, so the CMB, the CNB and the BAO follow from the standard Einstein-Boltzmann equations with the UM-2 particle content.

# 3 Space, coordinates and the periodic hidden direction

## 3.1 The coordinates (AUTHOR)

The author's coordinates are $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (the author's notebook: `X = {x0[], x1[], ..., x7[]}`), with

- $x_0$: the hidden space-like direction, $z \equiv 6Hx_0$ takes every real value;
- $x_1, x_2, x_3$: ordinary 3-space;
- $x_4$: the time;
- $x_5, x_6, x_7$: three extra time-like directions.

$H > 0$ is the author's constant. Frame indices $a = 0, \ldots, 7$ are aligned with $x_0, \ldots, x_7$, and

$$\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1).$$

## 3.2 The hidden coordinate $y$, the fold and the tips (DERIVED)

On every interval of $z$ on which $\sin z \neq 0$ define

$$y = \frac{\ln|\sin z|}{6H} \in (-\infty, 0], \qquad |\sin z| = e^{6Hy}, \qquad dy = \cot z\, dx_0.$$

One period $0 < z < \pi$ of $x_0$ (length $\ell_0 = \pi/(6H)$) consists of two halves, $0 < z < \pi/2$ and $\pi/2 < z < \pi$, joined at the **fold** $z = \pi/2$ ($y = 0$, where $\cos z = 0$). Each half ends at a **tip** ($z \to 0^+$ or $z \to \pi^-$, $y \to -\infty$), at infinite proper distance from the fold. As in the Kohn-Sham record of this repository the tip is replaced by a regular cutoff at $y = -L$ (ASSUMED), with

$$\eta \equiv HL = 3.$$

The fold reflection

$$R: \quad z \mapsto \pi - z \qquad (x_0 \mapsto \ell_0 - x_0)$$

maps the two halves onto each other; modes of a field on one period are even or odd under $R$. Every function of $y(z)$ is periodic in $x_0$ with period $\ell_0 = \pi/(6H)$.

# 4 The metric as a power series

The metric is a series of terms of decreasing magnitude,

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \ldots, \qquad \epsilon \sim 10^{-5},$$

where $\epsilon$ is the size of the CMB temperature anisotropy (STANDARD).

## 4.1 Order 0: the corrected primordial metric (AUTHOR coordinates, DERIVED properties)

In the author's chart, with two general functions of the time $a_4(x_4)$ and $b_4(x_4) > 0$,

$$\bar{g} = \mathrm{diag}\Big(e^{-2a_4} b_4 \cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big),$$

where $(\sin^2 z)^{1/6}$ is the positive branch. This is the metric of the Revision record of 2026-10-16 (version 2), whose sources are all diagonal. The notebook metric

$$g^{\mathrm{nb}} = \mathrm{diag}\big(\cot^2 z,\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ -1,\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6}\big),$$

to whose extra-time components the author's order of 2026-10-14 added the factor $b_4$, differs from it in $g_{00}$ and in the 3-space warp. Define

$$f = e^{-a_4} b_4^{1/2}, \qquad k = \frac{1}{f^2} = \frac{e^{2a_4}}{b_4}, \qquad A = \dot{a}_4, \qquad h = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4,$$

dots being $d/dx_4$. In the $y$ chart the metric is

$$d\bar{s}^2 = e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 + f^2\big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2)\big].$$

The block $\{x_0, x_5, x_6, x_7\}$ is $f^2$ times a four-dimensional space of constant curvature $-H^2$ and signature $(1,3)$; $f$ is its common scale factor ("the block"), $A$ is the observer's Hubble rate and $h < 0$ means that the extra times and $x_0$ deflate. The volume element and the vielbein are

$$\sqrt{|\bar{g}|} = e^{3a_4} f^4\, |\cot z|\, (\sin^2 z)^{1/4} = e^{3a_4} f^4 e^{3Hy} \Big|\frac{dy}{dx_0}\Big|,$$

$$e^0{}_{x_0} = f \cot z, \quad e^i{}_{x_i} = e^{a_4}\ (i = 1,2,3), \quad e^4{}_{x_4} = 1, \quad e^t{}_{x_t} = f(\sin^2 z)^{1/12}\ (t = 5,6,7).$$

**Curvature (DERIVED; sympy, identical in both charts, 25/25 checks).** The Riemann tensor is pair-diagonal, $R^{ab}{}_{cd} = 0$ unless $\{a,b\} = \{c,d\}$, and every sectional curvature $K_{ab} = R^{ab}{}_{ab}$ is one of

$$K_{ij} = A^2, \quad K_{i4} = A^2 + \dot{A}, \quad K_{iB} = Ah, \quad K_{4B} = h^2 + \dot{h}, \quad K_{BB'} = h^2 - H^2 k,$$

with $i,j \in \{1,2,3\}$ and $B, B' \in \{0,5,6,7\}$. None depends on $x_0$; the curvature is the same on both halves of every period.

## 4.2 Order 1: the observable perturbations and the hidden tower

Every field and every metric perturbation is expanded in the eigenfunctions $u_n(x_0)$ of its $x_0$ operator (section 6),

$$h^{(1)}_{MN}(x^\mu, x_0) = \sum_{n \geq 0} h^{(1,n)}_{MN}(x^\mu)\, u_n(x_0).$$

The $n = 0$ term has the flat profile $u_0 = \mathrm{const}$; in the Newtonian gauge its line element is

$$ds^2 = e^{2a_4}\big[(1 - 2\epsilon\Psi_N)\delta_{ij} + \epsilon h^{TT}_{ij}\big]dx_i dx_j - (1 + 2\epsilon\Phi)dx_4^2 + f^2(1 + 2\epsilon\sigma)\big[dy^2 - e^{2Hy}dx_t^2\big],$$

with the two scalar potentials $\Phi$, $\Psi_N$, the transverse-traceless tensor $h^{TT}_{ij}$ and the radion $\sigma$ (the relative fluctuation of the block size). The scalar and tensor sectors carry the CMB, the CNB and the BAO; the radion is massive (section 8.3) and does not propagate at cosmological distances. The terms $n \geq 1$ are Kaluza-Klein modes with masses above the gap $3H/(2f)$ (section 6). Order 2 contains the products of order-1 terms and the back-reaction of $x_0$-localised sources; it is not computed in this version (OPEN).

# 5 The action

$$S = S_{EL}[g] + S_{stab}[g] + S_\Psi[g,\Psi] + S_{gauge}[g,G,W,B] + S_f[g,\psi;G,W,B] \\ + S_{tip}[g;W,B,\Sigma,\psi] + S_{portal}[\Psi;\psi,G,W,B,\Sigma].$$

## 5.1 Einstein-Lovelock gravity (AUTHOR weights)

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\,\sqrt{|g|}\,\Big(\sum_{k=1}^{3}\alpha_k\,\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{a_1b_1\cdots a_kb_k}_{c_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

with the generalised Kronecker delta $\delta^{\cdots}_{\cdots}$ (the determinant of ordinary deltas). $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)} = R^2 - 4R_{ab}R^{ab} + R_{abcd}R^{abcd}$ (Gauss-Bonnet) and $\mathcal{L}_{(3)}$ is the cubic Lovelock density. The field equations are

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\,T^M{}_N, \qquad E_{(k)}{}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{Ma_1b_1\cdots a_kb_k}_{Nc_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

$E_{(1)}$ being the Einstein tensor. The author's weights are

$$\alpha_k = w_kH^{2-2k}, \qquad w = (w_1,w_2,w_3) = \Big(1,\frac{1}{32},\frac{1}{1152}\Big), \qquad \Lambda = -10H^2$$

(the author's notebook writes $\Lambda = 10$ with the opposite sign convention for $\Lambda$). UM-2 keeps these values.

**For a pair-diagonal curvature (DERIVED).** When $R^{ab}{}_{cd}$ is pair-diagonal (the order-0 metric),

$$\mathcal{L}_{(k)} = 2^k k! \sum_{\text{k disjoint pairs}}\ \prod_{\text{pairs}} K_{ab}, \qquad E_{(k)}{}^h{}_h = -2^{k-1}k! \sum_{\text{k disjoint pairs avoiding } h}\ \prod_{\text{pairs}} K_{ab},$$

which was checked against the Einstein tensor from the Ricci tensor ($k=1$), against the explicit Gauss-Bonnet tensor ($k=2$), against the Bianchi identity $\nabla_M E_{(k)}{}^M{}_N = 0$ ($k=1,2,3$), against the Euler-Lagrange equations of the action with a lapse ($k=1,2,3$) and against the record's equations (all exact).

## 5.2 The block stabiliser (ASSUMED)

A source term

$$S_{stab} = -\int d^8x\,\sqrt{|g|}\,\rho_b(f), \qquad \kappa\,\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8,$$

whose energy-momentum tensor is that of a vacuum energy of the block,

$$T^M{}_N[\text{stab}] = \text{diag}\big(p_0,p_3,p_3,p_3,-\rho_b,p_t,p_t,p_t\big), \qquad p_3 = -\rho_b, \qquad p_0 = p_t = +\rho_b.$$

It is conserved ($\dot\rho_b = -3A(\rho_b+p_3) - 4h(\rho_b+p_0) = -8h\rho_b$ gives $\rho_b \propto f^{-8}$), it is four-dimensionally Lorentz invariant ($p_3 = -\rho_b$) and isotropic in the block ($p_0 = p_t$), as the field equations require (section 8.1), and it does not depend on $x_0$. The $f^{-8}$ law is that of an eight-dimensional energy density whose only scale is the size of the block (a Casimir-type or flux energy). Its normalisation is fixed by the static block (section 8.2): $\kappa\rho_b^* = (26+4\sqrt{51})H^2$ at $f_* = k_*^{-1/2}$. Its microscopic origin is OPEN (a Casimir energy of the fields of the cell or a flux are natural candidates; neither was computed in UM-2).

## 5.3 The dirac16complex field

### 5.3.1 The Clifford algebra of the author (AUTHOR, DERIVED verification)

The author's real $16\times16$ matrices are built as follows ($h=1,2,3$; $p,q=1,\dots,4$; $\epsilon$ the Levi-Civita symbol; $I_n$ the identity):

$$Q_a(h,p,q) = \epsilon_{hpq4}, \quad Q_b(h,p,q) = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \quad s_h = Q_a - Q_b, \quad t_h = Q_a + Q_b,$$

$$\zeta_0 = I_8, \quad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}, \quad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad \zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6, \quad \sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix},$$

$$\tau_0 = I_8, \quad \tau_A = \zeta_A \ (A = 1..7), \quad \bar\tau_A = \sigma\,\tau_A^T\sigma, \qquad \gamma^{x_A} \equiv T16[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \quad (A = 0, \dots, 7).$$

They satisfy (DERIVED, exact; 31/31 checks)

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab} I_{16}, \qquad \tfrac{1}{2}(\tau_A\bar\tau_B + \tau_B\bar\tau_A) = \eta_{AB} I_8, \qquad \sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7.$$

The charge-conjugation matrix, the chirality, the Krein metric and the generators are

$$C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3} = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix}, \quad \Gamma = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7} = \mathrm{diag}(-I_8, I_8), \quad B = -iC\gamma^{x_4}, \quad S^{ab} = \tfrac{1}{4}[\gamma^a, \gamma^b],$$

with $C^T = C$, $C^2 = 1$, $(C\gamma^a)^T = -C\gamma^a$, $B^\dagger = B$, $B^2 = 1$ and $B$ of signature $(8,8)$. The 16-dimensional representation is irreducible under Pin(4,4) (commutant of dimension 1), the double cover of $O(4,4)$, and it splits under Spin(4,4), the double cover of $SO(4,4)$, into two inequivalent irreducible 8-dimensional representations, the eigenspaces $\Gamma = \mp 1$ (commutant of dimension 2, intertwiner of dimension 0).

### 5.3.2 The Lagrangian and the canonical spin connection

$\Psi$ has 16 complex anticommuting (Grassmann) components and $\bar\Psi = \Psi^\dagger C$. The Lagrangian density is

$$\mathcal{L}_\Psi = \sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\,\bar\Psi\Psi - \tfrac{\lambda}{2}(\bar\Psi\Psi)^2\Big],$$

$$\gamma^M = e_a{}^M\gamma^a, \quad D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \quad \Omega_M = \tfrac{1}{2}\,\omega_{Mab}S^{ab},$$

$$\omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}\,e_b{}^L\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b,$$

the canonical (metric-compatible, torsion-free) spin connection: it makes $\gamma^N$ covariantly constant, $\partial_M\gamma^N + \Gamma^N{}_{ML}\gamma^L + [\Omega_M, \gamma^N] = 0$ (DERIVED). For the order-0 metric (DERIVED, exact, for every $a_4$, $b_4$ and on both halves)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H = 0$ with constant $a_4$, $b_4$): the coupling of $\Psi$ to gravity is non-trivial. The field equation is

$$\gamma^M D_M\Psi = \big(m + \lambda\,\bar\Psi\Psi\big)\Psi.$$

The mass parameter is the author's $m = 3HM$; UM-2 uses $M = 1/3$, i.e. $m = H$ (ASSUMED, the value of the Kohn-Sham record). $\lambda$ (dimension mass$^{-6}$) is the contact self-coupling.

### 5.3.3 The two halves $\Psi_A$ and $\Psi_B$ and the fold parity (DERIVED)

$B$ commutes with every four-dimensional Lorentz generator $\gamma^a\gamma^b$ ($a, b \in \{x_1, .., x_4\}$) and with $C$, so

$$\Psi = \Psi_A + \Psi_B, \qquad B\Psi_A = -\Psi_A, \qquad B\Psi_B = +\Psi_B$$

splits $\Psi$ into two four-dimensionally Lorentz-invariant halves of 8 components, and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ does not mix them. In the good sector (no extra-time momenta) every interaction of UM-2 commutes with $B$ (DERIVED with the author's matrices: $B$ commutes with $C$, $\gamma^a$ and $C\gamma^a$ for $a = x_0, \dots, x_4$ and anticommutes with them for $a = x_5, x_6, x_7$), so the numbers $N_A$ and $N_B$ are separately conserved; extra-time momenta would mix the halves. Under the fold reflection $R$ the modes are even or odd; the bulk Lagrangian is $R$-symmetric.

## 5.4 The Standard-Model gauge fields (ASSUMED structure, STANDARD content)

The gauge group is $SU(3)_c \times SU(2)_L \times U(1)_Y$ with eight-dimensional gauge fields $G^\alpha_M$ ($\alpha = 1..8$), $W^a_M$ ($a = 1, 2, 3$) and $B_M$:

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\,g^{MP}g^{NQ}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^\alpha_{PQ} + \frac{1}{\hat g^2}W^a_{MN}W^a_{PQ} + \frac{1}{\hat g'^2}B_{MN}B_{PQ}\Big),$$

$$G^\alpha_{MN} = \partial_M G^\alpha_N - \partial_N G^\alpha_M + f^{\alpha\beta\gamma}G^\beta_M G^\gamma_N, \quad W^a_{MN} = \partial_M W^a_N - \partial_N W^a_M + \epsilon^{abc}W^b_M W^c_N, \quad B_{MN} = \partial_M B_N - \partial_N B_M.$$

The four-dimensional couplings are $g_s^2 = \hat{g}_s^2/V_\perp$, $g^2 = \hat{g}^2/V_\perp$, $g'^2 = \hat{g}'^2/V_\perp$, where $V_\perp$ is the proper volume of the block per unit 4-volume (section 7). In this geometric normalisation the couplings stand in front of the kinetic terms and not in the covariant derivatives; the dimensions are $[G_M] = [W_M] = [B_M] = 1$, $[\hat{g}^2] = -4$ (mass units), and the four-dimensional couplings are dimensionless. From here on $W^\pm$, $Z$, $A$ and every formula that shows $\hat{g}$, $\hat{g}'$ explicitly refer to the canonically normalised fields $W^a/\hat{g}$, $B/\hat{g}'$, $G^\alpha/\hat{g}_s$ (written with the same letters). The physical fields are

$$W_M^\pm = \frac{W_M^1 \mp i W_M^2}{\sqrt{2}}, \qquad Z_M = \cos\theta_W\, W_M^3 - \sin\theta_W\, B_M, \qquad A_M = \sin\theta_W\, W_M^3 + \cos\theta_W\, B_M, \qquad \tan\theta_W = \frac{g'}{g}.$$

**$W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$** ($[T_3, T_\pm] = \pm T_\pm$, $Y(W) = 0$; $W^-$ is the antiparticle of $W^+$); $Z$ and the photon $A$ are neutral (DERIVED). A fermion of weak isospin $T_3$ and hypercharge $Y$ couples to $A$ with $e(T_3 + Y) = eQ$, $e = gg'/\sqrt{g^2 + g'^2}$, and to $Z$ with $\sqrt{g^2 + g'^2}\,(T_3 - Q\sin^2\theta_W)$ (DERIVED). The non-abelian terms of $W^a_{MN} W^a_{PQ}$ contain the vertices $W^+W^-\gamma$ (strength $e$), $W^+W^-Z$ (strength $g\cos\theta_W$) and the quartic $W^+W^-W^+W^-$, $W^+W^-\gamma\gamma$, $W^+W^-Z\gamma$, $W^+W^-ZZ$; every vertex conserves electric charge, so a single $W^+$ or $W^-$ always comes with a charged partner.

## 5.5 The Standard-Model fermions (three generations; ASSUMED structure, INVENTED boundary assignment)

Each multiplet is an eight-dimensional spinor of the same Clifford algebra (16 components times its gauge indices). There are three generations, $i = 1, 2, 3$, with the same quantum numbers:

| field ($i = 1, 2, 3$) | members | $SU(3)_c$ | $SU(2)_L$ | $Y$ | electric charges $Q = T_3 + Y$ | 4D zero mode |
|---|---|---|---|---|---|---|
| $Q_i$ | $(u, d)_L$, $(c, s)_L$, $(t, b)_L$ | 3 | 2 | 1/6 | up-type 2/3, down-type $-1/3$ | $\Gamma_4 = +1$ |
| $u_{Ri}$ | $u_R$, $c_R$, $t_R$ | 3 | 1 | 2/3 | 2/3 | $\Gamma_4 = -1$ |
| $d_{Ri}$ | $d_R$, $s_R$, $b_R$ | 3 | 1 | $-1/3$ | $-1/3$ | $\Gamma_4 = -1$ |
| $L_i$ | $(\nu_e, e)_L$, $(\nu_\mu, \mu)_L$, $(\nu_\tau, \tau)_L$ | 1 | 2 | $-1/2$ | neutrino 0, charged lepton $-1$ | $\Gamma_4 = +1$ |
| $e_{Ri}$ | $e_R$, $\mu_R$, $\tau_R$ | 1 | 1 | $-1$ | $-1$ | $\Gamma_4 = -1$ |
| $\nu_{Ri}$ | three right-handed neutrinos | 1 | 1 | 0 | 0 | $\Gamma_4 = -1$ |

with the action

$$S_f = \sum_{i=1}^{3} \sum_{\psi} \int d^8x \sqrt{|g|} \Big[ \tfrac{1}{2} \big( \bar{\psi}_i \gamma^M D_M \psi_i - (D_M \bar{\psi}_i) \gamma^M \psi_i \big) - \mu_{\psi i} \bar{\psi}_i \psi_i \Big],$$

$$D_M = \partial_M + \Omega_M - i G^\alpha_M T^\alpha - i W^a_M \frac{\tau^a}{2} - i Y B_M,$$

$\psi \in \{Q, u_R, d_R, L, e_R, \nu_R\}$, the same canonical spin connection $\Omega_M$ as for $\Psi$ and a constant bulk mass $\mu_{\psi i}$ per multiplet. The bulk masses are diagonal in the generation index, and the doublet bulk mass is the same for the three generations of a sector (ASSUMED):

$$\mu_{Li} = \mu_L, \qquad \mu_{Qi} = \mu_Q \qquad (i = 1, 2, 3),$$

while the singlet bulk masses $\mu_{e_R i}$, $\mu_{\nu_R i}$, $\mu_{u_R i}$, $\mu_{d_R i}$ differ from generation to generation. The 4D chirality is $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$.

**Boundary conditions.** In 16-component form, with the selection projector

$$P_{sel} = \frac{1 + \Gamma_4 \gamma^{x_0}}{2}\, \frac{1 - i\gamma^{x_5}\gamma^{x_6}}{2},$$

which commutes with every four-dimensional Lorentz generator (DERIVED), the conditions at the fold ($y = 0$) and at the tip ($y = -L$) are

- doublets ($Q$, $L$; $\mu_\psi > 0$): fold $P_{sel}(1-\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi = 0$; tip $(1-\gamma^{x_0})\psi = 0$;
- singlets ($u_R$, $d_R$, $e_R$, $\nu_R$; $\mu_\psi < 0$): fold $P_{sel}(1+\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1-\gamma^{x_0})\psi = 0$; tip $(1+\gamma^{x_0})\psi = 0$.

These are self-adjoint boundary conditions of the Dirac operator (DERIVED). The result (DERIVED, section 6): every doublet field has exactly one four-dimensional Weyl zero mode with $\Gamma_4 = +1$ and every singlet field exactly one with $\Gamma_4 = -1$, all localised at the fold and all in the two blocks selected by $P_{sel}$, which lie in the $B = -1$ half. The zero-mode spectrum is three Standard-Model generations, each with its $\nu_R$. Writing every zero mode as a left-handed Weyl field (a right-handed field of hypercharge $Y$ as a left-handed one of hypercharge $-Y$), with $n_c$ colours and $n_w$ weak components, the anomaly coefficients of one generation are (DERIVED, exact fractions)

$$\mathcal{A}_{SU(3)^2U(1)} = \sum_{\text{triplets}} n_w Y = 0, \qquad \mathcal{A}_{SU(2)^2U(1)} = \sum_{\text{doublets}} n_c Y = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

$$\mathcal{A}_{U(1)^3} = \sum n_c n_w Y^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0, \qquad \mathcal{A}_{grav^2U(1)} = \sum n_c n_w Y = 1 - 2 + 1 - 1 + 1 = 0,$$

and $\mathcal{A}_{SU(3)^3}$ (left-handed minus right-handed triplets) $= 2 - 1 - 1 = 0$; the twelve $SU(2)$ doublets of three generations are an even number (no Witten anomaly). The anomalies cancel generation by generation; three lepton generations with a single quark generation would leave $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$, so the second and third quark generations are required together with the second and third lepton generations (DERIVED).

**QCD (STANDARD).** The gluons are the massless $x_0$ zero modes of $G^\alpha_M$. Below the QCD scale the quarks are confined into hadrons; the top quark decays (into $W^+b$) before it can form one. The baryons of the cosmology are the protons ($uud$, charge $+1$) and neutrons ($udd$, charge 0), whose masses $m_p = 938.3$ MeV and $m_n = 939.6$ MeV come from QCD; they are inputs of the cosmology. The QCD scale depends on the block through $g_s^2 = \hat{g}_s^2/V_\perp$.

## 5.6 The tip: electroweak symmetry breaking without a Higgs particle (INVENTED for this geometry)

At the regular tip a non-linear sigma field $\Sigma(x) \in SU(2)$, $\Sigma^\dagger\Sigma = 1$, transforming as $\Sigma \to U_L\,\Sigma\,U_Y^\dagger$ with $U_Y = \exp(i\theta\tau^3/2)$, carries the electroweak symmetry breaking:

$$S_{tip} = \int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \sum_{i,j=1}^{3}\big(\bar{L}_i\,\Sigma\,(\Lambda_\ell)_{ij}\,\ell_{Rj} + \bar{Q}_i\,\Sigma\,(\Lambda_q)_{ij}\,q_{Rj} + \text{h.c.}\big)\Big],$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW_\mu^a\frac{\tau^a}{2}\Sigma + iB_\mu\,\Sigma\frac{\tau^3}{2}, \qquad \ell_{Rj} = (\nu_{Rj}, e_{Rj}), \quad q_{Rj} = (u_{Rj}, d_{Rj}),$$

$$(\Lambda_\ell)_{ij} = \mathrm{diag}\big((\lambda_\nu)_{ij}, (\lambda_e)_{ij}\big), \qquad (\Lambda_q)_{ij} = \mathrm{diag}\big((\lambda_u)_{ij}, (\lambda_d)_{ij}\big),$$

with four complex $3\times 3$ tip coupling matrices $\lambda_\nu$, $\lambda_e$, $\lambda_u$, $\lambda_d$ (the diagonal acts on the isospin components: the upper component of $\Sigma^\dagger L_i$ couples to $\nu_R$, the lower one to $e_R$). $\Sigma$ has no radial mode, so there is no Higgs particle; its three would-be Goldstone modes become the longitudinal $W^+$, $W^-$ and $Z$. Every term is $SU(2)_L \times U(1)_Y$ invariant, the Yukawa terms for arbitrary complex generation matrices (DERIVED with random group elements and random matrices, deviation $1.1\times 10^{-14}$). In the unitary gauge $\Sigma = 1$ the first term is

$$-\frac{v_\Sigma^2}{8}\Big[\hat{g}^2\big(W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}\big) + (\hat{g}^2 + \hat{g}'^2)Z_\mu Z^\mu\Big] = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}^2\,W_\mu^+ W^{-\mu} + (\hat{g}^2 + \hat{g}'^2)Z_\mu Z^\mu\Big],$$

which gives $W^+$ and $W^-$ one common mass (they are particle and antiparticle), the $Z$ a mass larger by $1/\cos\theta_W$ at leading order (custodial symmetry), and leaves the photon massless. The second term mixes the doublet and singlet zero modes at the tip and gives the fermions their masses and their mixing (section 6.2).

## 5.7 The portal interactions of dirac16complex (INVENTED, gauge invariant)

$\Psi$ is a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$, so it couples to the Standard Model only through gauge-invariant composite operators. UM-2 contains two of them (the names I11 and I12 are those of version 1).

**I11: the vector contact portal.**

$$S_{I11} = \int d^8x\,\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar{\Psi}\gamma^M\Psi)\sum_{i=1}^{3}\sum_{\psi} q_\psi\,(i\bar{\psi_i}\gamma_M\psi_i),$$

summed over the multiplets of the three generations with charges $q_\psi$ (each term is separately gauge invariant, and generation blind); $\Lambda_\star$ is the common scale of the two bulk portals I11 and I12, and $c_V$ is dimensionless ($[\bar\Psi\gamma^M\Psi] = [\bar\psi\gamma_M\psi] = 7$). It couples both halves of $\Psi$, including the fold bands, to the fermion currents. To keep the sterile fold bands out of thermal equilibrium it must be weak: UM-2 sets $c_V = 0$ in the benchmark and computes only the linear response of $\Psi$ to it (section 9).

**I12: the gauge-kinetic scalar portal.**

$$S_{I12} = -\int d^8x\,\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_G\,G^\alpha_{MN}G^{\alpha MN} + c_W\,W^a_{MN}W^{aMN} + c_B\,B_{MN}B^{MN}\Big]$$

(field strengths in the geometric normalisation of section 5.4; $c_G, c_W, c_B$ dimensionless, since $[\bar\Psi\Psi] = 7$ and $[G_{MN}G^{MN}] = 4$). With $W^aW^a = 2W^+_{MN}W^{-MN} + W^3W^3 + \ldots$ and $W^3 = \cos\theta_W Z + \sin\theta_W A$, $B = -\sin\theta_W Z + \cos\theta_W A$, it couples $\bar\Psi\Psi$ to the pairs $W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$ and gluon pairs. The scalar density of every chiral zero mode vanishes identically (its lower component is zero), so **I12 does not touch the massless fold bands**; it couples the massive states of $\Psi$, in particular the dark matter, which it lets annihilate into gauge-boson pairs (section 11). In the QCD vacuum the gluon condensate $\langle G^\alpha G^\alpha\rangle$ makes I12 a uniform mass shift of $\Psi$.

**Removed in version 2: the chiral tip $Z$ portal.** Version 1 contained a third portal,

$$S_{I9} = \int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\,\big(c_Z + c'_Z\,\varsigma\big)\,(i\bar\Psi\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_\mu\Sigma\,\tau^3\big), \qquad P_+ = \frac{1+\Gamma_4}{2},$$

($\varsigma = \pm1$ distinguishing the two sides of a tip; in the unitary gauge $\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3) = \frac{v_\Sigma^2}{2}\sqrt{\hat g^2+\hat g'^2}\,Z_\mu$), whose even part $c_Z$ made the $\Psi_A$ fold band (two Weyl species) couple to the $Z$ like two neutrinos and so stand in for the missing $\nu_\mu$, $\nu_\tau$ of one generation. Its algebra is DERIVED and remains true: $P_+$ is a Hermitian projector commuting with the 4D Lorentz generators and with $C\gamma^\mu$; $\Gamma_4 = -B\gamma^{x_0}$, so $P_+ = (1 - B\gamma^{x_0})/2$, which is $(1+\gamma^{x_0})/2$ on $\Psi_A$ and $(1-\gamma^{x_0})/2$ on $\Psi_B$ and annihilates $\Psi_B$ at the tip (every orbital satisfies $(1-\gamma^{x_0})\Psi = 0$ there): **no $P_+$ tip current and no tip operator built on $\bar\Psi\Psi$ reaches $\Psi_B$** (a non-chiral or $P_-$ tip current would). With three neutrinos the portal has no role left, and both of its parts are bounded or disfavoured by the data (DERIVED, section 10.7):

- its even part gives the $Z$ an invisible width $N_\nu = 3 + 2r^2$ ($r$ the $Z$ coupling of the fold band relative to a neutrino's), bounded by the measured $N_\nu = 2.9963 \pm 0.0074$ to $r^2 < 0.0056$; even at that bound the fold band would stay in equilibrium down to 14.7 MeV and add

  $$\Delta N_{eff} = n_W\Big(\frac{10.75}{g_s}\Big)^{4/3}F_h, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4 = 1.0095$$

  ($n_W = 2$ Weyl species decoupling when the plasma has $g_s$ entropy degrees of freedom, here $g_s = 10.75$; $F_h$ corrects for the heating of the neutrinos by $e^+e^-$ annihilation, which an earlier-decoupled species does not receive: $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ of the common-temperature decoupling of section 9.4), i.e. $\Delta N_{eff} = 2.02$, far above the measured $N_{eff} = 2.99 \pm 0.17$ (Planck 2018) and $2.86 \pm 0.13$ (ACT DR6) (STANDARD);
- its oriented part $c'_Z$ (odd under the fold reflection) would let the heavy odd $\Psi_A$ states decay, $\Psi_A^{odd} \to \Psi_A^{fold} + Z$, and by the inverse decays and the exchange process $\Psi_A^{fold}Z \to \Psi_A^{fold}Z$ keep the fold band in equilibrium until the odd states or the $Z$ bosons disappear (at $g_s = 105.75$ or 86.25), adding $\Delta N_{eff} = 0.096$ to 0.126 (ESTIMATE), i.e. $N_{eff} = 3.14$ to 3.17, 2.1 to 2.4 standard deviations above the ACT DR6 value (disfavoured).

UM-2 therefore sets $c_Z = c'_Z = 0$ (ASSUMED). Consequences: $N_\nu = 3$ exactly; neither fold band is ever thermalised (no tip operator, zero scalar density for I12, $c_V = 0$ for I11); both halves keep an exact fold parity, so the lightest odd states of $\Psi_A$ and of $\Psi_B$ are both stable (section 11).

**Not included, and why.** A dipole portal $\bar\Psi S^{MN}\Psi B_{MN}$ would couple the fold bands to the photon and thermalise them (too much dark radiation). No gauge-invariant operator couples the neutral $\Psi$ to a single $W^+$ or $W^-$; $W^\pm$ enter only in pairs (I12) or with a charged fermion pair (the covariant derivatives of I11).

## 5.8 The catalogue of interactions

| # | interaction | fields | equation (vertex) | role in UM-2 |
|---|---|---|---|---|
| 1 | Einstein-Lovelock self-interaction | $g$ | $\sum_k \alpha_k \mathcal{L}_{(k)} - 2\Lambda$ | background, block dynamics, 4D Newton constant $\propto 1/G(k)$ |
| 2 | gravity of every field | $g$ with all | $\sqrt{\lvert g\rvert}$, $g^{MN}$, $\Omega_M$ in each action | universal coupling; $\Psi$ through the canonical spin connection |
| 3 | block stabiliser | $g$ | $-\sqrt{\lvert g\rvert}\rho_b(f)$ | static, stable block with attractive 4D gravity |
| 4 | $\Psi$ self-interaction | $\Psi$ | $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ | Kohn-Sham mean fields $M_{eff}$, $v$ |
| 5 | strong interaction | $G$, the six quarks | $-G^{\alpha}_{M}\,(i\bar{q}\gamma^{M}T^{\alpha}q)$, $f^{\alpha\beta\gamma}GGG$, $GGGG$ | confinement, $p$ and $n$ |
| 6 | electromagnetic interaction | $A$, charged fermions, $W^{\pm}$ | $-eQ\,A_M\,(i\bar{\psi}\gamma^{M}\psi)$; $W^{+}W^{-}\gamma$ (strength $e$) | CMB photons, recombination, Thomson drag |
| 7 | weak charged current | $W^{\pm}$, the three lepton and quark generations | $-\frac{g}{\sqrt{2}}W^{+}_{M}J^{M}_{CC}$ and h.c.; $J^{M}_{CC}=\sum_{\alpha} i\bar{\nu}_{\alpha}\gamma^{M}P_{+}\ell_{\alpha}$ $+\sum_{ab}V_{ab}\,i\bar{u}_{a}\gamma^{M}P_{+}d_{b}$; $\nu_{\alpha}=\sum_{i}U_{\alpha i}\nu_{i}$ | $\beta$ decay, $n\leftrightarrow p$, neutrinos produced and detected in flavour states, flavour conversion |
| 8 | weak neutral current | $Z$, all fermions | $-\sqrt{g^2+g'^2}\,Z_M$ $\times(T_3-Q\sin^2\theta_W)\,i\bar{\psi}\gamma^{M}\psi$, flavour diagonal in every basis | neutrino decoupling, $Z$ widths |
| 9 | gauge self-interactions | $W^{\pm}$, $Z$, $\gamma$ | $W^{+}W^{-}Z$, $W^{+}W^{-}W^{+}W^{-}$, $W^{+}W^{-}\gamma\gamma$, $W^{+}W^{-}Z\gamma$, $W^{+}W^{-}ZZ$ | electroweak consistency |
| 10 | $W^{\pm}$, $Z$ masses | $\Sigma$, $W^{a}$ ($W^{\pm}$, $W^{3}$), $B$ | $-\frac{v_{\Sigma}^2}{4}\mathrm{tr}(D\Sigma^{\dagger}D\Sigma)$ at the tip | $x_0$ profiles bent at the tip |
| 11 | fermion masses and mixing | $\Sigma$, $L_i$, $\ell_{Rj}$, $Q_i$, $q_{Rj}$ | $\sum_{ij}\left(\bar{L}_i\Sigma(\Lambda_{\ell})_{ij}\ell_{Rj}+\bar{Q}_i\Sigma(\Lambda_q)_{ij}q_{Rj}\right)$ at the tip | tip mixing of the chiral zero modes: masses, PMNS, CKM |
| 12 | neutrino forward scattering (7 and 8 at low energy) | $\nu$ with $e$, $n$, $\nu$ | $V_{CC}=\sqrt{2}G_F n_e$ on $\nu_e$; $V_{NC}$ flavour universal; $\nu\nu$ term | MSW conversion (Sun, Earth, supernovae), early-Universe potentials, fast flavour instability |
| 13 | $\Psi$ - fermion currents | $\Psi$, all fermions | I11 | bounded; Kohn-Sham response |
| 14 | $\Psi$ - gauge pairs | $\Psi$, $W^{+}W^{-}$, $ZZ$, $Z\gamma$, $\gamma\gamma$, $gg$ | I12 | dark-matter relic abundance; QCD-condensate mass shift |
| 15 | radion couplings | $\sigma$ with all massive fields | $\sigma\, m\,\frac{\partial \ln m}{\partial \ln f}\,\bar{\psi}\psi$, etc. | the masses track the block size |

# 6 The $x_0$ mass mechanism

The author's requirement: mass is given to every non-massless particle by the dependence of its wave function on the space coordinate $x_0$, periodic with period $\pi/(6H)$; localisation in $x_0$ implies, through $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$, a hidden momentum and hence an effective mass. UM-2 implements it exactly: every four-dimensional mass is an eigenvalue of the $x_0$ operator of its field, acting on wave functions of $y(z) = \ln|\sin z|/(6H)$, $z = 6Hx_0$, which are periodic in $x_0$ with period $\pi/(6H)$. Throughout this section the block is static ($f$ constant), $\xi = y/L \in [-1, 0]$ on each half, and eigenvalues are given in units of the proper half cell, $\Lambda = (m\,fL)^2$ for bosons and $e = \varepsilon\,fL$ for fermions.

## 6.1 Gauge fields and the graviton (DERIVED)

A four-dimensional component $A_\mu(x)\,u(y)$ of a gauge field (gauge $A_y = 0$) has the reduced action $\int_{-L}^{0} dy\, f^4 e^{3Hy}\big[(\partial_y u)^2/f^2 - m^2u^2\big]$ per half, hence the $x_0$ operator and boundary conditions

$$-\frac{1}{f^2}\,e^{-3Hy}\,\partial_y\big(e^{3Hy}\partial_y u\big) = m^2 u \quad\Longleftrightarrow\quad -e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda\,u,$$

$$\text{fold: } u'(0) = 0 \text{ (even)}, \quad u(0) = 0 \text{ (odd)}; \qquad \text{tip: } u'(-1) = \beta\,u(-1),$$

with $\beta = 0$ for the unbroken generators (photon, gluons) and $\beta > 0$ for the broken ones ($W^\pm$, $Z$; from the $\Sigma$ term). The transverse-traceless graviton obeys the same operator in Einstein gravity. With $u = e^{-3\eta\xi/2}\tilde{u}$ and $q^2 = \Lambda - 9\eta^2/4$ (primes denote $d/d\xi$) the solutions are elementary and the eigenvalue conditions are

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta\,q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0$$

($q = i\tilde{q}$ below the gap $9\eta^2/4$).

**Massless particles.** For $\beta = 0$ the flat profile $u = \text{const}$ is an exact eigenfunction with $\Lambda = 0$: the **graviton, the photon and the gluons are exactly massless**; their masslessness is protected by four-dimensional general covariance and gauge invariance. The excited levels are $\Lambda_n = 9\eta^2/4 + n^2\pi^2$ (even, $n \geq 1$) and the roots of $q\cot q = -3\eta/2$ (odd); for $\eta = 3$: $\Lambda_1 = 30.1196$, odd 27.0890, gap 20.25 (DERIVED, finite elements agree to $5\times10^{-10}$). The Kaluza-Klein scale is $m_{KK} = \sqrt{\Lambda_1}/(fL)$.

**$W^\pm$ and $Z$.** The $\Sigma$ term gives $\beta_W$ for $W^\pm$ (one value for both charges) and, by custodial symmetry, $\beta_Z = \beta_W/\cos^2\theta_W$, with $\theta_W$ the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2+\hat{g}'^2)$; at small $\beta$, $\Lambda \simeq \beta\cdot 3\eta e^{-3\eta}/(1-e^{-3\eta})$ and $m_Z/m_W = 1/\cos\theta_W$. Beyond leading order the bending of the profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ ($\Lambda_W$, $\Lambda_Z$ the lowest even eigenvalues for $\beta_W$ and $\beta_Z$) away from $\cos^2\theta_W$: the electroweak parameter

$$\rho = \frac{\Lambda_W}{\Lambda_Z\cos^2\theta_W}$$

exceeds 1. This tree-level custodial violation is what the measured $\rho_0 - 1 = 0.00038$ constrains ($\rho_0$ measures the violation beyond the Standard-Model loops). With the measured masses (STANDARD: $m_W = 80.3625$ GeV, $m_Z = 91.1879$ GeV, PDG 2026; $\rho_0 - 1 = 0.00038$), $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = \rho_0$ fixes the angle of the couplings, $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (the on-shell value $m_W^2/m_Z^2$ is 0.776663), and the fit gives (DERIVED)

$$\beta_W = 0.011912, \qquad \Lambda_W = 1.32149\times10^{-5}, \qquad m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W} = 121.3\ \text{TeV},$$

with both masses reproduced: $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W} = 91.1879$ GeV. A positive tip term cannot make $W^\pm$ heavier than the Dirichlet limit $\tanh\tilde{q} = 2\tilde{q}/(3\eta)$, $\Lambda_D = 0.010016$, i.e. $m_W \leq 0.01824\,m_{KK}$.

## 6.2 Fermions (DERIVED)

The eight-dimensional Dirac equation with the canonical spin connection reduces EXACTLY (all eight blocks, general $a_4(x_4)$, $b_4(x_4)$, $y$-dependent mass $M(y)$ and potential $v(y)$; DERIVED with the author's matrices) under

$$\Psi = e^{ikx_1}\,e^{-\frac{3}{2}Hy}\,e^{a_4/2}\,b_4^{-1}\,V_\beta\,\chi(y, x_4)$$

to $i\partial_4\chi = h_j\chi$ ($k$ the momentum of the mode along $x_1$) with the $2\times 2$ block Hamiltonian

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

where $V_\beta$ ($\beta = 0..7$) are the columns of the block basis of the commuting operators $J = \gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$, $K_1 = \gamma^{x_2}\gamma^{x_3}$, $K_2 = \gamma^{x_5}\gamma^{x_6}$ with eigenvalues $(j, is_2, is_3)$. In every block $\gamma^{x_0} = \sigma_3$, $B = js_2$, $C = s_2\sigma_2$ and $\Gamma_4 = -js_2\sigma_3$. The factor $e^{-\frac{3}{2}Hy}$ removes the $\frac{3}{2}(H/f)\gamma^{x_0}$ of the spin connection, and $e^{a_4/2}b_4^{-1}$ removes its $\gamma^{x_4}$ part. For a constant mass $\mu$ and $k = 0$ the general solution of $h_j\chi = 0$ is $\chi = (A\,e^{f\mu y}, B\,e^{-f\mu y})$; the boundary conditions of section 5.5 select one component. With $c = f\mu L$:

- doublet zero mode $\chi = (e^{c\xi}, 0)$, $\Gamma_4 = +1$, localised at the fold for $c > 0$;
- singlet zero mode $\chi = (0, e^{-c\xi})$, $\Gamma_4 = -1$, localised at the fold for $c < 0$;
- at $k \neq 0$: $h_j(e^{c\xi}, 0) = je^{-a_4}k\,(e^{c\xi}, 0)$ exactly: the zero modes are massless four-dimensional particles with $\varepsilon = e^{-a_4}|k|$ (physical momentum).

**Masses from the tip: one generation.** The tip coupling turns the tip conditions into the self-adjoint mixing condition (DERIVED)

$$b_L(-1) = \kappa_f\,b_R(-1), \qquad a_R(-1) = -\kappa_f\,a_L(-1), \qquad \chi = (a, ib),$$

with $\kappa_f$ the dimensionless tip mixing. At $k = 0$ the doublet ($c_L > 0$) and singlet ($c_R < 0$) solutions of $a' = c\,a - e\,b$, $b' = e\,a - c\,b$ (block $j = +1$, eigenvalue $e = \varepsilon fL$) that satisfy the fold conditions reach the tip with

$$(a_L, b_L) = (C_L - c_L s_L,\ -s_L e), \qquad (a_R, b_R) = (s_R e,\ C_R + c_R s_R), \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

so the four-dimensional mass is the lowest root of the exact $x_0$ eigenvalue equation (DERIVED; direct ODE shooting of the coupled system agrees to $6\times 10^{-10}$)

$$D(e) = e^2 s_L s_R - \kappa_f^2\,(C_L - c_L s_L)(C_R + c_R s_R) = 0,$$

whose first-order solution is

$$e = \kappa_f\,\zeta(c_L)\,\zeta(-c_R), \qquad \zeta(c)^2 = \frac{c\,e^{-c}}{\sinh c} \simeq 2c\,e^{-2c} \quad (c > 0),$$

i.e. the tip mixing times the probability densities of the two chiral wave functions at the tip ($c_L > 0$ and $c_R < 0$: both zero modes are localised at the fold).

**Masses and mixing from the tip: three generations.** In each sector (neutrinos, charged leptons, up-type quarks, down-type quarks) the tip coupling is a $3\times 3$ matrix $\kappa$ (the dimensionless form of $\lambda_\nu$, $\lambda_e$, $\lambda_u$, $\lambda_d$), and the conditions hold for the vectors of boundary values over the generations:

$$b_L(-1) = \kappa\,b_R(-1), \qquad a_R(-1) = -\kappa^\dagger a_L(-1),$$

which is self-adjoint for every complex $\kappa$ (DERIVED: the boundary form vanishes identically, sympy, 42 real variables). With the doublet amplitudes $\mathcal{A}_\alpha$ and the singlet amplitudes $\mathcal{B}_i$ of the solutions above, $\mathrm{diag}(b_{L\alpha})\mathcal{A} = \kappa\,\mathrm{diag}(b_{Ri})\mathcal{B}$ and $\mathrm{diag}(a_{Ri})\mathcal{B} = -\kappa^\dagger\mathrm{diag}(a_{L\alpha})\mathcal{A}$, hence

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\,\mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0:$$

the masses are the roots $e$ of the determinant of this matrix, and $\mathcal{A}$ is the left-handed flavour content of each mass state. If the three doublets of a sector share one bulk mass ($a_{L\alpha} = a_L$, $b_{L\alpha} = b_L$) and $\kappa$ is unitary, the matrix is $\kappa\big[b_L + a_L\,\mathrm{diag}(b_{Ri}/a_{Ri})\big]\kappa^\dagger$, so

$$\det\big[\,\ldots\,\big] = \prod_{i=1}^{3}\frac{b_L a_{Ri} + a_L b_{Ri}}{a_{Ri}}, \qquad \mathcal{A}^{(i)} = \kappa\,\hat{e}_i, \qquad \mathcal{B}^{(i)} \propto \hat{e}_i$$

($\hat{e}_i$ the unit vectors): each factor is the one-generation condition with $\kappa_f = 1$ ($b_L a_R + a_L b_R = -D(e)$), so the masses are the one-generation roots, the left-handed rotation of the sector is its tip matrix (so that the mixing

matrices are $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$), and the right-handed rotations are trivial (DERIVED with mpmath at 50 digits: the roots of the $3 \times 3$ determinant equal the one-generation roots, both refined in the same computation, to $3.0 \times 10^{-42}$ in all four sectors; the masses stored by the mass computation, found with a stopping tolerance near $10^{-11}$, agree with them to $4.5 \times 10^{-11}$; the null vectors equal the columns of $\kappa$ to $1.1 \times 10^{-16}$; for a random complex non-unitary $\kappa$ the exact roots agree with the singular values of $\zeta(c_L)\,\kappa\,\mathrm{diag}(\zeta(-c_{Ri}))$ to $3.5 \times 10^{-5}$).

**The UM-2 choice (ASSUMED).** One doublet bulk mass per sector, common to the three generations ($c_L = f\mu_L L$ for the leptons, $c_Q = f\mu_Q L$ for the quarks), fixed by the heaviest fermion of the sector with $c_R = -c_L$ ($\tau$) and $c_R = -c_Q$ ($t$); unitary tip matrices

$$\kappa_e = \kappa_u = 1, \qquad \kappa_\nu = U_{PMNS}, \qquad \kappa_d = V_{CKM}$$

in the basis in which the singlet bulk masses are diagonal. The measured masses (STANDARD, the PDG 2025 values: $m_e$, $m_\mu$, $m_\tau$; $m_u$, $m_d$, $m_s$ in $\overline{MS}$ at 2 GeV, $m_c(m_c)$, $m_b(m_b)$, the directly measured $m_t$, used as tree-level targets) and the neutrino masses of section 10.1 then require (DERIVED; exact roots; the first-order formula agrees to $1.1 \times 10^{-3}$ for the top quark and better than $3 \times 10^{-4}$ for every other fermion)

| fermion | mass | doublet $c$ | singlet $c_R$ | $\mu_R/H$ | singlet width $fL/(2\lvert c_R\rvert)$ | $e = m\,fL$ |
|---|---|---|---|---|---|---|
| $\nu_1$ | 1.000 meV | 5.9529 | $-35.0452$ | $-54.194$ | $1.27 \times 10^{-22}$ m | $4.524 \times 10^{-17}$ |
| $\nu_2$ | 8.712 meV | 5.9529 | $-32.8481$ | $-50.796$ | $1.36 \times 10^{-22}$ m | $3.941 \times 10^{-16}$ |
| $\nu_3$ | 50.140 meV | 5.9529 | $-31.0702$ | $-48.047$ | $1.44 \times 10^{-22}$ m | $2.268 \times 10^{-15}$ |
| $e$ | 0.51100 MeV | 5.9529 | $-14.5539$ | $-22.506$ | $3.07 \times 10^{-22}$ m | $2.312 \times 10^{-8}$ |
| $\mu$ | 105.658 MeV | 5.9529 | $-8.9809$ | $-13.888$ | $4.97 \times 10^{-22}$ m | $4.779 \times 10^{-6}$ |
| $\tau$ | 1776.93 MeV | 5.9529 | $-5.9529$ | $-9.205$ | $7.50 \times 10^{-22}$ m | $8.038 \times 10^{-5}$ |
| $u$ | 2.16 MeV | 3.3823 | $-15.4300$ | $-23.861$ | $2.89 \times 10^{-22}$ m | $9.771 \times 10^{-8}$ |
| $c$ | 1.2730 GeV | 3.3823 | $-8.7683$ | $-13.559$ | $5.09 \times 10^{-22}$ m | $5.758 \times 10^{-5}$ |
| $t$ | 172.56 GeV | 3.3823 | $-3.3823$ | $-5.230$ | $1.32 \times 10^{-21}$ m | $7.806 \times 10^{-3}$ |
| $d$ | 4.70 MeV | 3.3823 | $-14.6258$ | $-22.617$ | $3.05 \times 10^{-22}$ m | $2.126 \times 10^{-7}$ |
| $s$ | 93.5 MeV | 3.3823 | $-11.5158$ | $-17.808$ | $3.88 \times 10^{-22}$ m | $4.230 \times 10^{-6}$ |
| $b$ | 4.183 GeV | 3.3823 | $-7.5005$ | $-11.599$ | $5.95 \times 10^{-22}$ m | $1.892 \times 10^{-4}$ |

The doublet bulk masses are $\mu_L = 9.2055\,H$ (width $7.50 \times 10^{-22}$ m) and $\mu_Q = 5.2304\,H$ (width $1.32 \times 10^{-21}$ m). Order-one bulk masses give the whole hierarchy, because the tip densities depend exponentially on $c$; within a sector the lighter fermion has the sharper singlet localisation. The neutrinos are Dirac fermions; their right-handed partners $\nu_{Ri}$ are gauge singlets that are never thermalised. (Version 1 gave every fermion its own $c_L = -c_R$; with three generations that rule would give the members of one doublet different bulk masses, against the single bulk mass of a multiplet, and flavour-changing Kaluza-Klein couplings, below.)

**Mixing matrices (STANDARD parametrisation, measured values).** Both tip matrices are written as

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \qquad J = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta,$$

with, for the PMNS matrix (NuFIT 6.0, normal ordering), $\sin^2\theta_{12}=0.308$, $\sin^2\theta_{23}=0.470$, $\sin^2\theta_{13}=0.02215$, $\delta=212^\circ$ ($\theta_{12}=33.71^\circ$, $\theta_{23}=43.28^\circ$, $\theta_{13}=8.56^\circ$; $J=-0.01777$), and for the CKM matrix (PDG global fit) $s_{12}=0.22501$, $s_{13}=0.003732$, $s_{23}=0.04183$, $\delta=1.147$ ($J=3.117\times10^{-5}$). Both are unitary to $1.1\times10^{-16}$ (DERIVED). The magnitudes are

$$|U|=\begin{pmatrix}0.8226 & 0.5488 & 0.1488\\ 0.3351 & 0.6543 & 0.6779\\ 0.4594 & 0.5203 & 0.7199\end{pmatrix},\qquad |V|=\begin{pmatrix}0.97435 & 0.22501 & 0.00373\\ 0.22488 & 0.97349 & 0.04183\\ 0.00858 & 0.04111 & 0.99912\end{pmatrix}.$$

**No flavour-changing neutral currents at tree level (DERIVED).** The first even Kaluza-Klein gauge mode couples to a zero mode of localisation $c$ through the overlap $\int|\chi_c|^2u_1\,d\xi$ ($u_1$ normalised like the zero mode); relative to the zero mode this is 0.14434 for every left-handed lepton and $-0.30066$ for every left-handed quark, because the doublets share one bulk mass, and the right-handed rotations are trivial, so in the mass basis the largest off-diagonal coupling is $10^{-17}$ (round-off). With different doublet bulk masses the PMNS rotation would create off-diagonal couplings up to 0.033. The $W$ and $Z$ zero modes couple to every generation with the same strength (lepton universality); the PMNS and CKM matrices carry the whole flavour structure of the charged currents. Charged-lepton flavour violation through the neutrino masses, $\mathrm{BR}(\mu\to e\gamma)=\frac{3\alpha}{32\pi}\big|\sum_i U^*_{\mu i}U_{ei}\,m_i^2/m_W^2\big|^2=2.8\times10^{-55}$, is unobservable.

## 6.3 The quanta of dirac16complex (DERIVED)

$\Psi$ has the same block Hamiltonian with $M=m$ and the boundary conditions fold $(1-\gamma^{x_0})\Psi=0$ (even) or $(1+\gamma^{x_0})\Psi=0$ (odd) and tip $(1-\gamma^{x_0})\Psi=0$; both parities are modes of the full period. With $c_\psi=fmL$ the exact $k=0$ spectrum is

- even: the zero mode $\chi=(e^{c_\psi\xi},0)$, $\varepsilon=0$, in all eight blocks, and $\varepsilon=\pm\sqrt{m^2+(n\pi/(fL))^2}$;
- odd: $\varepsilon=\pm\sqrt{m^2+p^2}$ with $\tan(p\,fL)=-p/m$.

The **fold band** $\varepsilon=e^{-a_4}|k|$ is massless: 8 states per 3-momentum, 4 in $\Psi_A$ (two Weyl species with $\Gamma_4=+1$) and 4 in $\Psi_B$ (two Weyl species with $\Gamma_4=-1$). The **bulk band** is massive; at the stabilised block ($f_*=0.21556$, $c_\psi=0.64667$) its lowest level is the odd level $\varepsilon_{DM}=3.10220\,m$ ($p\,fL=1.89901$, $p=2.93661\,m$), and the lowest even bulk level is $4.95998\,m$. The odd level is the dark-matter candidate (section 11).

## 6.4 What the uncertainty relation does and does not bound (DERIVED)

In the flat measure $u=e^{-3\eta\xi/2}\tilde u$ the exact identity $e^{3\eta\xi}u'^2=\tilde u'^2+\frac94\eta^2\tilde u^2-\frac32\eta\,(\tilde u^2)'$ turns every bosonic eigenvalue into

$$(m\,fL)^2=\langle p^2\rangle+\frac{9\eta^2}{4}-\frac{3\eta}{2}\,\varrho(\text{fold})+\Big(\frac{3\eta}{2}+\beta\Big)\varrho(\text{tip}),\qquad \langle p^2\rangle=\frac{\int\tilde u'^2}{\int\tilde u^2},\quad \varrho=\frac{\tilde u^2}{\int\tilde u^2},$$

and every fermionic one into $(\varepsilon fL)^2=\langle|\chi'|^2\rangle+c^2-c\,[\chi^\dagger\sigma_3\chi]_{-1}^0/\int|\chi|^2$, with $\langle p^2\rangle=\langle|\chi'|^2\rangle$ and $\varrho=|\chi|^2/\int|\chi|^2$ (integrals over $\xi\in[-1,0]$, primes $d/d\xi$). The proper uncertainty relation bounds only the hidden kinetic energy. On the cell, which ends at the tips, it reads

$$\sqrt{\langle p^2\rangle}\,\Delta\xi\ge\frac{1-\varrho(\text{tip})}{2},\qquad \Delta\xi^2=\int_{-1}^{0}\xi^2\varrho\,d\xi,$$

since $1=\int\varrho\,d\xi=\varrho(\text{tip})-\int\xi\varrho'\,d\xi$ (the fold end $\xi=0$ contributes nothing) and the Cauchy-Schwarz inequality bounds $|\int\xi\varrho'\,d\xi|$ by $2\Delta\xi\sqrt{\langle p^2\rangle}$; $\Delta s=fL\,\Delta\xi$ is the root-mean-square proper distance from the fold. The familiar $\langle p^2\rangle\ge1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip (it can hold otherwise: the first even Kaluza-Klein mode, with $\varrho(\text{tip})=0.655$, has $\langle p^2\rangle\Delta\xi^2=4.59$); the photon zero mode has 0.497, the broad fold zero mode of $\Psi$ ($\varrho(\text{tip})=0.489$) only 0.097. The relation with the tip term is a theorem; it is also checked numerically for each of the 21 computed states (the zero mode of the photon, gluons and graviton, $W^\pm$, $Z$, the first even and odd Kaluza-Klein modes, the two doublet zero modes $c_L$ and $c_Q$, the twelve singlet zero modes of the fermions, the fold zero mode of $\Psi$ and the dark matter; DERIVED). Three classes follow:

- **massless**: profiles flat relative to the geometry ($u=$ const or $h\chi=0$): the fold term cancels the kinetic energy and the warp exactly; graviton, photon, gluons and the $\Psi$ fold band. They are localised at the fold ($\Delta s=1.4\times10^{-21}$ m for the photon) and yet massless;

- **heavy**: the $x_0$ dependence fills the cell. For the gauge fields and the graviton every odd mode and every excitation has $m \geq 3H/(2f) = 99.5$ TeV (the gap theorem of the boson operator: one even level below the gap for every $\beta \geq 0$, no odd level); the $\Psi$ excitations start at its lowest odd level, $3.10m = 44.35$ TeV (the dark matter), since the Dirac operator has no warp gap. For the computed states of this class the author's estimate $\hbar c/(2\Delta s)$, with $\Delta s$ the root-mean-square proper distance from the fold, lies below the mass (by factors 2.3 to 7.5, the dark matter included); this is observed for these states, not derived for all;
- **light**: the $x_0$ dependence that costs mass sits at the tip, where the probability is tiny: $W^{\pm}$, $Z$ and the twelve fermions. Their masses lie $3 \times 10^{-3}$ (the top quark) to $10^{-17}$ ($\nu_1$) below $\hbar c/(2\Delta s)$ ($\Delta s$ of the broader of the fermion's two zero modes; $10^{-3}$ for $W^{\pm}$), and a sharper localisation of either zero mode gives a LIGHTER fermion ($d\ln m_f/d\ln c = -2.9$ to $-34.5$).

# 7 Dimensional reduction and the four-dimensional constants

The good sector (no dependence on $x_5, x_6, x_7$) is kept (ASSUMED): modes with extra-time momentum $q$ have $\omega^2 = M^2 + k^2 - q^2/f^2$ and are tachyonic for $q > fM$ (OPEN). The extra times are compact with coordinate volume $v_t$ (ASSUMED). Integrating over one period of $x_0$ (both halves) and the extra times, the proper block volume per unit 4-volume is

$$V_\perp = 2\int_{-L}^{0} dy\, f^4 e^{3Hy}\, v_t = \frac{2f^4 v_t\,(1-e^{-3\eta})}{3H},$$

and the reduced action of section 8.5 gives

$$\bar{M}_{Pl}^2 = \frac{G(k)\,V_\perp}{\kappa} = \frac{1}{8\pi G_N}, \qquad g^2 = \frac{\hat{g}^2}{V_\perp}, \quad g'^2 = \frac{\hat{g}'^2}{V_\perp}, \quad g_s^2 = \frac{\hat{g}_s^2}{V_\perp}.$$

($\bar{M}_{Pl}$ is the reduced Planck mass). Every four-dimensional constant depends on the block: $\alpha$ and the gauge couplings $\propto f^{-4}$, $G_N \propto f^{+2.05}$ ($d\ln G_N/d\ln f = -4 + 2kG'(k)/G(k)$ at $k_*$: the Lovelock factor varies too), Kaluza-Klein masses $\propto f^{-1}$, $m_W \propto f^{-0.50}$, the fermion masses as $f^{-6.8}$ (top quark) to $f^{-41}$ ($\nu_1$) ($m_e \propto f^{-20.5}$), $M_{DM} \propto f^{-0.77}$ (DERIVED). The masses are constant only because the block is stabilised.

# 8 Gravity: the background, the stabilised block, the deflating extra times

## 8.1 The background field equations (DERIVED)

For the order-0 metric every off-diagonal component of every $E_{(k)}$ vanishes and the hidden and extra-time components coincide, so the source must be diagonal, independent of $x_0$ and block-isotropic:

$$T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t), \qquad p_0 = p_t.$$

The three independent components are the $x_4$ constraint (first order in time derivatives), the 3-space component and the block component; the last two are linear in $(\dot{A}, \dot{h})$. With the pairing formula of section 5.1, the sectional curvatures of section 4.1, the author's weights and $\Lambda = -10H^2$ they are ($A = \dot{a}_4$, $h = \dot{f}/f$, $k = 1/f^2$, dots $d/dx_4$; DERIVED, sympy)

$$\begin{aligned} x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\ & - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\ & - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\ & = -\kappa\rho \end{aligned}$$

$$
\begin{aligned}
x_1:\quad & \Big[-\frac{h^4}{8H^4}-\frac{Ah^3}{2H^4}-\frac{3h^2}{2H^2}+\frac{kh^2}{4H^2}-\frac{Ah}{H^2}+\frac{Ahk}{2H^2} \\
& \quad -2+\frac{3k}{2}-\frac{k^2}{8}\Big]\dot{A} \\
& +\Big[-\frac{Ah^3}{2H^4}-\frac{3A^2h^2}{4H^4}-\frac{3h^2}{2H^2}-\frac{3Ah}{H^2}+\frac{Ahk}{2H^2}-\frac{A^2}{2H^2} \\
& \quad +\frac{kA^2}{4H^2}-4+\frac{3k}{2}\Big]\dot{h} \\
& -\frac{Ah^5}{2H^4}-\frac{19A^2h^4}{16H^4}-\frac{A^3h^3}{2H^4}-\frac{15h^4}{8H^2}-\frac{6Ah^3}{H^2}+\frac{Akh^3}{2H^2} \\
& -\frac{17A^2h^2}{4H^2}+\frac{7kA^2h^2}{8H^2}-\frac{hA^3}{H^2}+\frac{hkA^3}{2H^2}-10h^2+\frac{9kh^2}{4} \\
& -8Ah+3Ahk-3A^2+\frac{9kA^2}{4}-\frac{3A^2k^2}{16}-10H^2 \\
& +6kH^2-\frac{3H^2k^2}{8} \\
& =\kappa p_3 \\
x_0:\quad & \Big[-\frac{3Ah^3}{8H^4}-\frac{9A^2h^2}{16H^4}-\frac{9h^2}{8H^2}-\frac{9Ah}{4H^2}+\frac{3Ahk}{8H^2}-\frac{3A^2}{8H^2} \\
& \quad +\frac{3kA^2}{16H^2}-3+\frac{9k}{8}\Big]\dot{A} \\
& +\Big[-\frac{9A^2h^2}{16H^4}-\frac{3hA^3}{8H^4}-\frac{3h^2}{8H^2}-\frac{9Ah}{4H^2}-\frac{9A^2}{8H^2}+\frac{3kA^2}{16H^2} \\
& \quad -3+\frac{3k}{8}\Big]\dot{h} \\
& -\frac{9A^2h^4}{16H^4}-\frac{17A^3h^3}{16H^4}-\frac{9A^4h^2}{16H^4}-\frac{3h^4}{8H^2}-\frac{27Ah^3}{8H^2}-\frac{45A^2h^2}{8H^2} \\
& +\frac{3kA^2h^2}{16H^2}-\frac{27hA^3}{8H^2}+\frac{9hkA^3}{16H^2}-\frac{3A^4}{8H^2}+\frac{3kA^4}{16H^2}-6h^2 \\
& +\frac{3kh^2}{8}-9Ah+\frac{9Ahk}{8}-6A^2+\frac{9kA^2}{4}-10H^2 \\
& +3kH^2 \\
& =\kappa\bar{p}
\end{aligned}
$$

with $\bar{p}=(p_0+3p_t)/4$, and the source is conserved,

$$\dot{\rho}=-3A(\rho+p_3)-h(\rho+p_0)-3h(\rho+p_t)=-3A(\rho+p_3)-4h(\rho+\bar{p}).$$

A source with $p_0\neq p_t$ (the Kohn-Sham states of $\Psi$ and the matter at the fold have it; the dark matter has $p_0/\rho=0.772$, $p_t=0$) acts on the common block scale $f$ through the block-averaged pressure $\bar{p}=(p_0+3p_t)/4$ (the variation of the action with respect to $f$), while $p_0-p_t$ drives a block-shape mode that the order-0 metric does not contain (OPEN); for matter its size relative to the stabiliser is of order $(A/H)^2$, $10^{-92}$ today. To first order about the static block of section 8.2 with block-averaged matter sources ($A^2$ counts as first order: the $x_4$ constraint gives $3G(k_*)A^2=\kappa\delta\rho$, because $\frac{d}{dk}[6H^2k(1-2w_2k)+\kappa\rho_b]$ vanishes at $k_*$), the equations give (DERIVED)

$$\delta\dot{A}=J_{Ak}\,\delta k+s_{A3}\,\kappa\delta p_3+s_{Ap}\,\kappa\delta\bar{p}+q_A\,\kappa\delta\rho,$$

$$\delta\dot{h}=J_{hk}\,\delta k+s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar{p}+q_h\,\kappa\delta\rho,\qquad \delta\dot{k}=-2k_*\,\delta h,$$

$$\delta\ddot{k}+\omega_r^2\,\delta k=-2k_*\big(s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar{p}+q_h\,\kappa\delta\rho\big),\qquad \omega_r^2=2k_*J_{hk},$$

with $J_{Ak}=\frac{6(238-19\sqrt{51})}{2249}H^2=0.272956H^2$, $J_{hk}=\frac{4063-485\sqrt{51}}{2249}H^2=0.266522H^2$, $s_{A3}=\frac{2283}{4498}-\frac{162\sqrt{51}}{2249}=-0.006853$, $s_{Ap}=\frac{28\sqrt{51}-114}{2249}=0.038221$, $q_A=\frac{135021}{265382}-\frac{10308\sqrt{51}}{132691}=-0.045997$, $s_{h3}=\frac{21\sqrt{51}}{2249}-\frac{171}{4498}=0.028666$, $s_{hp}=\frac{239\sqrt{51}}{6747}-\frac{485}{2249}=0.037320$, $q_h=\frac{57}{4498}-\frac{7\sqrt{51}}{2249}=-0.009555$. In the adiabatic response ($\delta\ddot{k}=0$) the block shifts by $\delta k=(-0.107556\,\kappa\delta p_3-0.140028\,\kappa\delta\bar{p}+0.035852\,\kappa\delta\rho)/H^2$ and 3-space obeys exactly $\delta\dot{A}=-\kappa(\delta\rho+\delta p_3)/(2G(k_*))$: the four-dimensional Friedmann acceleration equation with $8\pi G_N=\kappa/(G(k_*)V_\perp)$.

## 8.2 The static block and attractive gravity with the author's weights (DERIVED)

For a static block ($h = 0$, $k$ constant) and $\dot{A} = 0$ the components are

$$x_4: \quad -3G(k)A^2 + 6H^2k(1-2w_2k) + \Lambda + \kappa\rho = 0, \qquad x_1: \quad \text{the same with } -\kappa p_3 \text{ for } \kappa\rho,$$

$$x_0: \quad 6(12w_2k-1)A^2 - \frac{12(w_2-18w_3k)}{H^2}A^4 + 3H^2k + \Lambda - \kappa p_0 = 0, \qquad G(k) = 1 - 24w_2k + 72w_3k^2.$$

$G(k)$ is the coefficient of the four-dimensional Einstein-Hilbert term (section 8.5): gravity is attractive for $G > 0$. Without sources the only static block with $A = 0$ (and $k \neq 0$) is $k_0 = 1/(4w_2) = 8$ with $\Lambda_0 = -24H^2$ and $G(k_0) = -5 + 9w_3/(2w_2^2) = -1$: with the author's weights the sourceless block has REPULSIVE gravity, and the author's $\Lambda = -10H^2$ is not $\Lambda_0$. For the author's weights $G(k) = (k^2 - 12k + 16)/16$, positive only for $k < 6 - 2\sqrt{5}$ or $k > 6 + 2\sqrt{5} = 10.47$.

With the block stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) and $A = 0$ the two conditions require

$$\kappa\rho_b = 10H^2 - 6H^2k + \tfrac{3}{8}H^2k^2 = \kappa p_0 = 3H^2k - 10H^2 \quad \Longrightarrow \quad k_\pm = 12 \pm \tfrac{4}{3}\sqrt{51}.$$

The upper root is the UM-2 vacuum:

$$k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219, \qquad f_* = k_*^{-1/2} = 0.215556,$$

$$G(k_*) = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0, \qquad \kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2,$$

with attractive four-dimensional gravity and the author's weights and $\Lambda$ unchanged. The lower root $k_- = 2.4781$ has $G = -0.4748$.

## 8.3 Stability: the radion (DERIVED)

Linearising the 3-space and block components (solved for $\dot{A}$, $\dot{h}$) with $\dot{k} = -2kh$ and $\rho_b \propto k^4$ about $k_*$ gives the eigenvalues $0, \pm i\,\omega_r$ with

$$\frac{\omega_r^2}{H^2} = \frac{31552}{2249} - \frac{2416\sqrt{51}}{6747} = 11.4721, \qquad \omega_r = 3.38705\,H:$$

the block is stable and oscillates; the zero eigenvalue is the direction of a small four-dimensional cosmological term. The radion (the quantum of block-size oscillations) has the mass $m_r = 3.387H = 48.4$ TeV. Stability holds for each of the stabiliser laws $p_0 = \nu\rho_b$, $\rho_b \propto k^{2(1+\nu)}$, computed at $\nu = 1/4, 1/2, 3/4, 1, 5/4, 3/2, 2, 3$ (radion masses $3.33H$ to $4.10H$).

## 8.4 The deflating extra times (DERIVED, full nonlinear equations)

The full nonlinear equations (all three components, the $x_4$ constraint preserved to $6 \times 10^{-10}$) were integrated from an expanding 3-space ($A$ from the constraint) and a block LARGER than the stabilised one ($k_0 < k_*$, i.e. $f_0 > f_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for $k_0$ from 1.5 to 20. Every start with $k_0$ above the separatrix $k_c(h_0) = 2.6983, 2.5991, 2.5301$ (for $h_0 = -H/2, -H/3, -H/10$; just above the unstable static block $k_- = 2.4781$, whose stable manifold the separatrix is) ends at $k_*$: $x_5, x_6, x_7$ and $x_0$ deflate (at rates up to $|h| = 2.63H$ in the stored runs) by up to the factor $f(k_c)/f_* = (k_*/k_c)^{1/2} = 2.917$, reach $k_*$ for the first time within $x_4 < 1/H$, and then oscillate about $k_*$ with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 = 0.667$ to $0.670$ late, $2/3$ for dust). Starts with $k_0 < k_c$ (still larger blocks) become singular within $x_4 = 0.2/H$ to $1.4/H$ or admit no expanding initial data. The extra times therefore deflate exponentially only for a short primordial era and are then frozen, which is what the constancy of the masses requires (section 7).

## 8.5 The four-dimensional effective gravity (DERIVED)

On a product of a four-dimensional metric with the static block (constant curvature $K = -H^2k$), the Lovelock densities factorise, $\mathcal{L}_{(p)} = \sum_{r+s=p} \frac{p!}{r!\,s!}\mathcal{L}^{(4)}_{(r)}\mathcal{L}^{(int)}_{(s)}$ ($p = 1, 2, 3$ the Lovelock order) with $\mathcal{L}^{(int)}_{(s)} = (1, 12K, 24K^2, 0)$ for

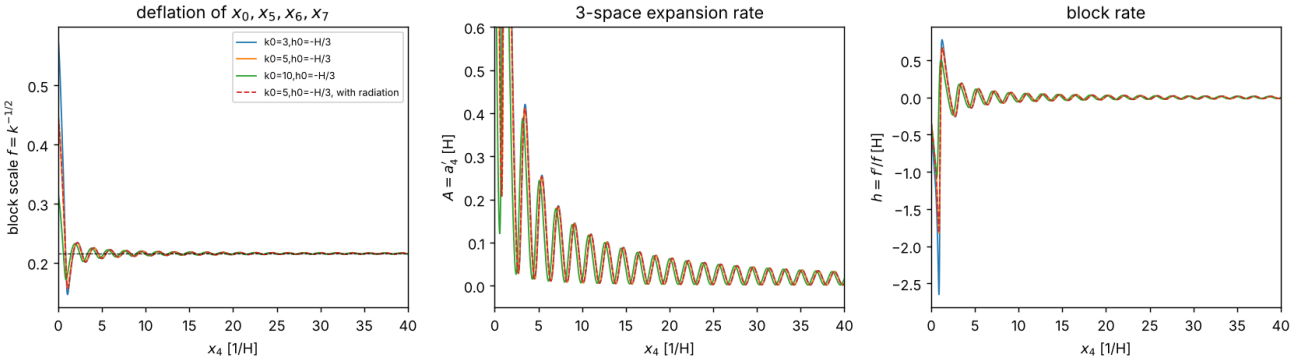



Figure 1: Deflation of the extra times into the stabilised block (full nonlinear Einstein-Lovelock equations with the author's weights).

$s = 0, 1, 2, 3$, and $\mathcal{L}^{(4)}_{(3)} \equiv 0$ in four dimensions while $\mathcal{L}^{(4)}_{(2)}$ (the four-dimensional Gauss-Bonnet density) is topological. The reduced Lagrangian (gravity plus the stabiliser, in units of $1/(2\kappa)$) is

$$\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)\, R^{(4)} + (\text{topological}) - 2\, G(k)\, \Lambda_4, \qquad \Lambda_4 = \frac{6H^2 k(1-2w_2 k) + \Lambda + \kappa\rho_b}{G(k)},$$

so the four-dimensional gravity of UM-2 is EXACTLY general relativity with Newton's constant $8\pi G_N = \kappa/(G(k_*)V_\perp)$ and the cosmological term $\Lambda_4$, which vanishes at the static point by construction. The observed dark energy is a small residual of this balance, $\Lambda_4/H^2 \sim 10^{-92}$: tuned (the cosmological-constant problem, OPEN).

# 9 dirac16complex in the cell: Kohn-Sham description

$\Psi$ is quantised canonically with $x_4$ as the time, $\{\Psi_r(x), \Psi^\dagger_s(x')\} = B_{rs}\, \delta^7(x-x')/\sqrt{|g|}$ on a slice ($r, s$ spinor indices); the indefinite Krein metric $B$ is handled in the good sector by counting the positive branch of every block as particles (the record's convention, ASSUMED). The interacting many-body state is described by Kohn-Sham theory: orbitals of $h_j$ with the mean fields

$$M_{eff}(y) = m + \tfrac{15}{16}\lambda S(y) + \delta M_{portal}(y), \qquad v(y) = -\tfrac{1}{16}\lambda\, n(y) + v_{portal}(y),$$

$$e_{int} = \tfrac{15}{32}\lambda S^2 - \tfrac{1}{32}\lambda n^2, \qquad E_{KS} = \sum_o g f_o \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{int}\, dy,$$

where $n$ and $S = \bar\Psi\Psi$ are the proper densities, $g = 4r_3(|\vec n|^2)$ the degeneracy of the momentum shell $\vec k = \Delta k\, \vec n$ ($\vec n \in \mathbb{Z}^3$, $r_3(|\vec n|^2)$ the number of representations of $|\vec n|^2$ as a sum of three squares, $\Delta k = 0.25H$), $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f^4$ (at $a_4 = 0$) the normalisation volume of a cell, of dimension length$^6$ (the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1-e^{-3\eta})/(3H)$), $f_o$ the occupations (aufbau or Mermin), and the portal mean fields come from I11 (the densities of the fermions of the three generations) and I12 (the gluon condensate). The energy-momentum tensor of a Kohn-Sham state,

$$\rho = \sum_o w g f_o \varepsilon_o\, n_o - e_{int}, \qquad p_3 = \sum_o w g f_o\, \tfrac{e^{-a_4|k|}}{3} t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum_o w g f_o \big[(\varepsilon_o - v) n_o - M_{eff} s_o - e^{-a_4}|k| t_o\big] + e_{int},$$

with $n_o = P\chi^\dagger_o \chi_o$, $s_o = P\chi^\dagger_o j\sigma_2 \chi_o$, $t_o = P\chi^\dagger_o j\sigma_3\chi_o$, $P = e^{-3Hy}/\mathrm{Vol}_7$ and $w = 1/2$ (the densities are $n = \sum_o w g f_o n_o$ and $S = \sum_o w g f_o s_o$), is the source of $\Psi$ in the gravitational equations; its block-averaged pressure $(p_0 + 3p_t)/4$ pushes on the block (section 8.1).

# 10 Three neutrino flavours: masses, mixing and flavour conversion (DERIVED)

## 10.1 Masses and mixing

The neutrinos are Dirac fermions whose masses are $x_0$ eigenvalues (section 6.2: the lepton doublet zero mode $c_L = 5.9529$ mixed at the tip with the singlet zero modes $c_R = -35.0452, -32.8481, -31.0702$). The lightest mass is not measured; UM-2 takes (ASSUMED) the normal ordering, the best fit of NuFIT 6.0 (the inverted ordering is disfavoured by $\Delta\chi^2 = 6.1$), and

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV},$$

with $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV$^2$ and $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV$^2$ (NuFIT 6.0). The sum $\sum m_\nu = 0.0599$ eV lies above the minimum 0.0588 eV of the normal ordering and below the DESI DR2 bound 0.064 eV (95 %, ΛCDM), which allows $m_1 < 4.1$ meV. The mixing is the neutrino tip matrix, $U = \kappa_\nu$, with the NuFIT 6.0 angles and phase of section 6.2. A neutrino produced by the charged current with the charged lepton $\ell_\alpha$ is the flavour state $|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |\nu_i\rangle$ ($\alpha = e, \mu, \tau$).

## 10.2 The flavour evolution

At energies far below $m_W$ the charged and neutral currents (catalogue rows 7 and 8) act as the four-fermion interaction with $G_F/\sqrt{2} = g^2/(8m_W^2)$ (STANDARD; UM-2 uses the measured $G_F = 1.1663788 \times 10^{-5}$ GeV$^{-2}$). Coherent forward scattering in matter then adds potentials to the free propagation, and the flavour amplitudes $\psi_\alpha$ of a neutrino of energy $E$ evolve as (STANDARD)

$$i\frac{d\psi}{dx_4} = \mathcal{H}\psi, \qquad \mathcal{H} = \frac{1}{2E}\, U\, \mathrm{diag}(m_1^2, m_2^2, m_3^2)\, U^\dagger + \mathrm{diag}(V_{CC}, 0, 0) + V_{NC}\, 1,$$

$$V_{CC} = \sqrt{2}\, G_F n_e, \qquad V_{NC} = -\frac{G_F n_n}{\sqrt{2}}$$

($n_e$, $n_n$ the electron and neutron densities; $V_{NC}$ is flavour universal and drops out of the probabilities); for antineutrinos $U \to U^*$ and $V_{CC}, V_{NC} \to -V_{CC}, -V_{NC}$. In vacuum

$$P(\nu_\alpha \to \nu_\beta) = \Big| \sum_i U_{\beta i} U_{\alpha i}^*\, e^{-i m_i^2 L/(2E)} \Big|^2, \qquad P(\nu_\alpha \to \nu_\beta) = \bar{P}(\nu_\beta \to \nu_\alpha) \text{ (CPT)},$$

$$P(\nu_\mu \to \nu_e) - P(\nu_e \to \nu_\mu) = -16 J \sin\Delta_{21} \sin\Delta_{31} \sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E}.$$

In a dense neutrino gas (the early Universe, a supernova core) the state of the neutrinos of momentum $\vec{p}$ is a $3 \times 3$ density matrix $\rho_{\vec{p}}$ (and $\bar{\rho}_{\vec{p}}$ for antineutrinos), which obeys the quantum kinetic equation (STANDARD)

$$i(\partial_{x_4} + \vec{v}_p \cdot \nabla)\rho_{\vec{p}} = \big[\Omega_{\vec{p}} + \mathrm{diag}(V_{CC}, 0, 0) + V_{\nu\nu}, \rho_{\vec{p}}\big] + i\,\mathcal{C}[\rho], \qquad \Omega_{\vec{p}} = \frac{U\, \mathrm{diag}(m_i^2)\, U^\dagger}{2p},$$

$$V_{\nu\nu} = \sqrt{2}\, G_F \int \frac{d^3q}{(2\pi)^3}\, (1 - \vec{v}_p \cdot \vec{v}_q)\big(\rho_{\vec{q}} - \bar{\rho}_{\vec{q}}\big),$$

with $\vec{v}_p = \vec{p}/p$ and $\mathcal{C}$ the collision term.

## 10.3 Vacuum, matter, the Sun and accelerators

The computed evolution (exact matrix exponentials, the layer products for varying densities) reproduces the vacuum formula to $3.9 \times 10^{-15}$, unitarity, CPT and the T-odd difference (DERIVED). Representative results (DERIVED; normal ordering, $\delta = 212°$):

- reactor $\bar{\nu}_e$ of 4 MeV survive with $\bar{P}_{ee} = 0.918$ at 1.65 km, 0.226 at 52.5 km and 0.267 at 180 km; atmospheric $\nu_\mu$ have their first survival minimum, 0.006, at $L/E = 1516$ km/GeV;

- the MSW resonance of the 1-3 sector, $2EV_{CC} = \Delta m^2_{ee}\cos 2\theta_{13}$ with $\Delta m^2_{ee} = c^2_{12}\Delta m^2_{31} + s^2_{12}\Delta m^2_{32}$, lies at 11.135 GeV in rock of 2.8 g/cm$^3$ ($Y_e = 0.5$; the exact three-flavour resonance $|U^m_{e3}|^2 = 1/2$ is at 11.133 GeV); that of the 1-2 sector at the centre of the Sun, $2EV_{CC}c^2_{13} = \Delta m^2_{21}\cos 2\theta_{12}$, at 1.93 MeV;
- $\nu_\mu \to \nu_e$ appearance at the long-baseline experiments (matter of constant density):

| experiment | $L$ | $E$ | $\rho$ (g/cm$^3$) | $P(\nu_\mu \to \nu_e)$ | $\bar{P}(\bar{\nu}_\mu \to \bar{\nu}_e)$ | vacuum $P(\nu_\mu \to \nu_e)$ |
|---|---|---|---|---|---|---|
| T2K | 295 km | 0.6 GeV | 2.6 | 0.0517 | 0.0313 | 0.0478 |
| NOvA | 810 km | 2.0 GeV | 2.84 | 0.0481 | 0.0236 | 0.0404 |
| DUNE | 1300 km | 2.5 GeV | 2.848 | 0.0701 | 0.0195 | 0.0491 |

- solar $\nu_e$ arrive at the Earth as an incoherent mixture of mass states, with the adiabatic survival probability

$$P_{ee} = c^4_{13}\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta^m_{12}\Big) + s^4_{13}, \qquad \cos 2\theta^m_{12} = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}c^2_{13}}{\Delta m^2_{21}},$$

evaluated at the production point in the electron density $n_e = N_A \min(245\, e^{-10.54\, r/R_\odot}, 100)$ cm$^{-3}$ (ASSUMED profile): $P_{ee} = 0.541$ (pp, 0.3 MeV), 0.520 ($^7$Be), 0.498 (pep), 0.377, 0.319, 0.306 ($^8$B at 5, 10, 15 MeV); the numerical three-flavour evolution through the solar density agrees with the adiabatic formula to $1.3 \times 10^{-3}$: the conversion is vacuum dominated at low energy and matter dominated (MSW) for $^8$B.

## 10.4 The early Universe and $N_{eff}$

The $e^\pm$ plasma gives the electron neutrinos a thermal potential (STANDARD)

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_F p}{3m^2_W}\,\rho_{e^\pm}, \qquad \rho_{e^\pm} = \frac{7\pi^2}{60}T^4,$$

which suppresses the mixing until it falls below $\Delta m^2\cos 2\theta/(2p)$; at the mean momentum $p = 3.15T$ the atmospheric oscillations start at $T = 4.99$ MeV and the solar ones at $T = 2.39$ MeV (DERIVED): flavour conversion is active while the neutrinos decouple. Decoupling is computed from the temperature equations (STANDARD form, Escudero 2020; MeV units)

$$\frac{dT_\gamma}{dx_4} = -\frac{4H\rho_\gamma + 3H(\rho_e + p_e) + 3HT_\gamma\frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta x_4} + 2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma\frac{d^2P_{int}}{dT^2_\gamma}}, \qquad \frac{dT_{\nu_\alpha}}{dx_4} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4} = \frac{G^2_F}{\pi^5}\Big[4(g^2_{eL} + g^2_{eR})F(T_\gamma, T_{\nu_e}) + 2F(T_{\nu_\mu}, T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4} = \frac{G^2_F}{\pi^5}\Big[4(g^2_{\mu L} + g^2_{\mu R})F(T_\gamma, T_{\nu_\mu}) - F(T_{\nu_\mu}, T_{\nu_e})\Big],$$

$$F(T_1, T_2) = 32 f_a(T^9_1 - T^9_2) + 56 f_s T^4_1 T^4_2(T_1 - T_2), \qquad f_a = 0.884, \quad f_s = 0.829,$$

with $H = \sqrt{8\pi\rho/3}/M_{Pl}$, $\rho_\gamma = \pi^2T^4_\gamma/15$, $\rho_{\nu_\alpha} = 7\pi^2T^4_{\nu_\alpha}/120$ per flavour, the $e^\pm$ energy density and pressure with the electron mass, $\nu_\tau$ following $\nu_\mu$, the effective couplings $g_{eL} = 0.727$, $g_{eR} = 0.233$, $g_{\mu L} = -0.273$, $g_{\mu R} = 0.233$ (or the UM-2 tree-level values $\frac{1}{2} + s^2$, $s^2$, $-\frac{1}{2} + s^2$, $s^2$ with $s^2 = 0.22363$), and the finite-temperature QED pressure of the $e^\pm\gamma$ plasma, $P_{int} = P_2 + P_3$,

$$P_2 = -\frac{e^2T^2}{12\pi^2}J_1 - \frac{e^2}{8\pi^4}J^2_1 + \frac{e^2m^2_e}{16\pi^4}J_2, \qquad P_3 = \frac{e^3T}{12\pi^4}I^{3/2}, \qquad \rho_{int} = -P_{int} + T\frac{dP_{int}}{dT},$$

$$J_1 = \int_0^\infty dp\,\frac{p^2}{E}n_D, \qquad J_2 = \int_0^\infty\int_0^\infty dp\,dq\,\frac{pq}{E_pE_q}\ln\Big|\frac{p+q}{p-q}\Big|n_D(p)n_D(q),$$

$$I = \int_0^\infty dp\,\frac{p^2 + E^2}{E}n_D, \qquad n_D = \frac{2}{e^{E/T} + 1} \qquad (E = \sqrt{p^2 + m^2_e}).$$

Then

$$N_{eff} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big].$$

With efficient oscillations the three flavours share one temperature, $dT_\nu/dx_4 = -HT_\nu + (\delta\rho_{\nu_e}/\delta x_4 + 2\delta\rho_{\nu_\mu}/\delta x_4)/(3\,d\rho_\nu/dT_\nu)$. The method reproduces the published values of the same method (DERIVED: instantaneous decoupling 3.0002, with the QED corrections 3.0108 and 3.0098; Fermi-Dirac energy transfer 3.0393, with QED 3.0495 and 3.0485; published 3.000, 3.011, 3.010, 3.039, 3.049). For UM-2 it gives $N_{eff} = 3.0476$ with separate $\nu_e$ and $\nu_{\mu,\tau}$ temperatures (published with separate temperatures: 3.048) and 3.0485 with one common temperature; $T_{\nu_e}/T_\gamma = 0.71732$, $T_{\nu_\mu}/T_\gamma = 0.71621$, the comoving neutrino temperatures being raised by 0.265 % ($\nu_e$) and 0.109 % ($\nu_{\mu,\tau}$) by $e^+e^-$ annihilation. The plasma at MeV temperatures is that of the three-flavour Standard Model and there is no extra radiation (neither fold band is ever thermalised), so UM-2 adopts the precision Standard-Model value with oscillations and the electron mass in the rates,

$$N_{eff} = 3.0432 \pm 0.0002 \qquad \text{(Cielo, Escudero, Mangano and Pisanti 2023).}$$

## 10.5 The cosmic neutrino background

After decoupling the flavour states lose their coherence: the CNB of UM-2 is an incoherent mixture of the three MASS eigenstates with the temperature and densities (DERIVED; instantaneous decoupling; $T_{CMB} = 2.7255$ K, STANDARD)

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_{CMB} = 1.9454 \text{ K}, \qquad n_i = \frac{3\zeta(3)}{4\pi^2}\,2T_\nu^3 = 112.0 \text{ cm}^{-3} \text{ per mass state } (\nu+\bar\nu), \qquad 336 \text{ cm}^{-3} \text{ in all,}$$

$$\Omega_\nu h^2 = \frac{1}{\rho_c/h^2}\sum_i 2\int \frac{d^3p}{(2\pi)^3}\frac{\sqrt{p^2+m_i^2}}{e^{p/T_\nu}+1} = 6.382\times10^{-4} \qquad (\rho_c/h^2 = 1.05368\times10^4 \text{ eV/cm}^3)$$

($6.451\times10^{-4}$ with the heated spectrum of $N_{eff} = 3.0432$: the degeneracy $N_{eff}/3$ of each state corresponds to a Fermi-Dirac spectrum of temperature $(N_{eff}/3)^{1/4}T_\nu$, so the non-relativistic density is multiplied by $(N_{eff}/3)^{3/4}$). The mean velocities today are $\langle v\rangle = \int_0^\infty p^2(p/E_i)f\,dp/\int_0^\infty p^2 f\,dp$ with $f = 1/(e^{p/T_\nu}+1)$, $E_i = \sqrt{p^2+m_i^2}$:

| state | mass | mean velocity today | $\nu_e$ content $\lvert U_{ei}\rvert^2$ | $\nu_\mu$ content | $\nu_\tau$ content |
|---|---|---|---|---|---|
| $\nu_1$ | 1.000 meV | $0.435\,c$ | 0.677 | 0.112 | 0.211 |
| $\nu_2$ | 8.712 meV | $0.0604\,c$ | 0.301 | 0.428 | 0.271 |
| $\nu_3$ | 50.140 meV | $0.0105\,c$ | 0.022 | 0.460 | 0.518 |

## 10.6 Supernovae and the dense neutrino gas

**MSW in the envelope.** In the envelope, with the profile $\rho = 10^4\,(10^9\,\text{cm}/r)^3$ g/cm$^3$ and $Y_e = 0.5$ (ASSUMED), a resonance lies at $n_e = \Delta m^2\cos 2\theta/(2\sqrt{2}\,G_F E)$; it is crossed adiabatically when the adiabaticity

$$\gamma = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta\,|d\ln n_e/dr|} \gg 1, \qquad P_{LZ} = e^{-\pi\gamma/2}$$

being the Landau-Zener probability of a jump (two-flavour conditions: $\Delta m^2_{31}$ and $\theta_{13}$ for the H resonance, $\Delta m^2_{21}$ and $\theta_{12}$ for the L resonance, without the factor $c^2_{13}$ of the solar condition). For 20 MeV the 1-3 (H) resonance lies at 1573 g/cm$^3$ ($r = 1.85\times10^4$ km) with $\gamma = 178$ ($P_{LZ} = 2.7\times10^{-122}$): adiabatic. For the normal ordering a $\nu_e$ from the core leaves the star as $\nu_3$ and an $\bar\nu_e$ as $\bar\nu_1$: $P(\nu_e\to\nu_e) = |U_{e3}|^2 = 0.022$, $\bar P(\bar\nu_e\to\bar\nu_e) = |U_{e1}|^2 = 0.677$ (without collective effects and Earth matter; DERIVED). For neutrinospheres with luminosities $L = 6, 6, 4\times10^{52}$ erg/s and mean energies $\langle E\rangle = 11, 14, 15.5$ MeV for $\nu_e$, $\bar\nu_e$, $\nu_x$ (ASSUMED), number fluxes $N\propto L/\langle E\rangle$, the mean energies at the Earth are $\langle E\rangle_{\nu_e} = [PN_e\langle E\rangle_e + (1-P)N_x\langle E\rangle_x]/[PN_e + (1-P)N_x] = 15.29$ MeV and, with $\bar P$, 14.34 MeV for $\bar\nu_e$.

**The fast flavour instability.** Near the neutrinosphere the neutrino gas is dense enough for $V_{\nu\nu}$ to dominate. For an axially symmetric gas with the electron-lepton-number (ELN) angular distribution $G(u) = \sqrt{2}\,G_F[n_{\nu_e}g_{\nu_e}(u) - n_{\bar\nu_e}g_{\bar\nu_e}(u)]$ ($u = \cos\vartheta$, $\int_{-1}^1 g\,du/2 = 1$), the linearised equations for the flavour coherence $S_u$ give plane-wave modes

$e^{-i(\Omega x_4 - K x_3)}$ along the symmetry axis $x_3$ with $\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$, $\Lambda^\mu = \int du/2\, G\, v^\mu$, $v = (1, u)$, and the dispersion relation (STANDARD; $\eta^{\mu\nu} = \mathrm{diag}(1, -1)$ on $(x_4, x_3)$ in this analysis)

$$\det \Pi = 0, \qquad \Pi^{\mu\nu} = \eta^{\mu\nu} + \int_{-1}^{1} \frac{du}{2}\, G(u)\, \frac{v^\mu v^\nu}{\omega - ku},$$

whose axisymmetric branch is $(1 + J_0)(-1 + J_2) - J_1^2 = 0$, $J_n = \int du/2\, G\, u^n/(\omega - ku)$, and whose azimuthal branch is $-1 + \int du/2\, G(1-u^2)/(2(\omega - ku)) = 0$; $\mathrm{Im}\,\omega > 0$ is an instability. In units of $\mu = \sqrt{2} G_F n_{\nu_e}$ (DERIVED): with an ELN crossing (forward-peaked antineutrinos, $\alpha = n_{\bar\nu_e}/n_{\nu_e} = 0.8$) the largest growth rate is $0.088\,\mu$ (homogeneous mode $0.018\,\mu$); with linear distributions ($g_{\nu_e} = 1 + 0.2u$, $g_{\bar\nu_e} = 1 + 0.6u$, $\alpha = 0.9$) it is $0.098\,\mu$ although the homogeneous mode is stable; without a crossing no mode grows, as the general theorem (Morinaga 2022) states. At $n_{\nu_e} = 10^{32}$ to $10^{33}$ cm$^{-3}$ the instability grows by $e$ in $6 \times 10^{-10}$ to $6 \times 10^{-11}$ s over 18 to 1.8 cm, a million times faster than the hydrodynamics.

**Flavour conversion and the explosions.** Conversion moves energy from the electron-flavour neutrinos, which heat the matter behind the shock through charged-current absorption, to the heavy-lepton neutrinos, which do not; a schematic estimate of the heating, $q \propto N_e \langle E^2\rangle_e + N_{\bar e}\langle E^2\rangle_{\bar e}$ with $\langle E^2\rangle = \frac{5}{4}\langle E\rangle^2$ (Fermi-Dirac-like spectra with the pinching parameter 3), compares full flavour equipartition, $q' = \frac{1}{3}(N_e\langle E^2\rangle_e + 2N_x\langle E^2\rangle_x) + \frac{1}{3}(N_{\bar e}\langle E^2\rangle_{\bar e} + 2N_x\langle E^2\rangle_x)$, and ELN-conserving pair conversion of $\Delta = \frac{2}{3}(N_{\bar e} - N_x)$ pairs, $q'' = (1 - \Delta/N_e)N_e\langle E^2\rangle_e + (1 - \Delta/N_{\bar e})N_{\bar e}\langle E^2\rangle_{\bar e}$, with $q$: the factors 0.884 and 0.760 for the neutrinospheres above (DERIVED, schematic). In 195 collapse simulations (9 to 120 solar masses) with instantaneous flavour equipartition below a baryon density $\rho_c$, Gogilashvili and Tamborra (2026) find the fraction of failed explosions 25.6 % without conversion and 50.8, 77.4, 93.3, 96.4 % for $\rho_c = 10^9$, $10^{10}$, $10^{11}$, $10^{13}$ g/cm$^3$ (27.0 % and 47.6, 61.9, 73.9, 88.3 % weighted with the Salpeter mass function), most strongly between 16 and 30 solar masses, with lighter neutron stars from the successful explosions (STANDARD, cited).

### 10.7 The $Z$ boson: invisible width and the widths of the three generations

With three neutrinos and no tip $Z$ portal the invisible width counts

$$N_\nu \equiv \frac{\Gamma_{inv}(Z)}{\Gamma(Z \to \nu\bar\nu)} = 3,$$

against the measured $2.9963 \pm 0.0074$ (PDG 2025; 0.5 standard deviations); a fold band of $Z$ coupling $r$ times a neutrino's would add $2r^2$, bounded by $r^2 < 0.0056$ (section 5.7). The tree-level partial widths (STANDARD)

$$\Gamma(Z \to f\bar f) = \frac{N_c G_F m_Z^3}{6\sqrt{2}\pi}\left(g_V^2 + g_A^2\right), \qquad g_V = T_3 - 2Q\sin^2\theta_W, \quad g_A = T_3,$$

with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$ give (DERIVED) 165.9 MeV per neutrino (497.7 MeV invisible; measured 499.1 MeV), 83.4 MeV per charged lepton, 285.4 MeV for each of $u$, $c$ and 367.9 MeV for each of $d$, $s$, $b$ (the top quark is too heavy), a hadronic width of 1674.4 MeV (1737.3 MeV with the QCD factor $1 + \alpha_s/\pi$, $\alpha_s = 0.1180$) and a total of 2422.3 MeV (2485.2 MeV with the QCD factor) against the measured 2495.5 MeV; $R_\ell = \Gamma_{had}/\Gamma_{\ell\ell} = 20.83$ against 20.77. The remaining differences are the size of the electroweak loop corrections, which UM-2 does not compute (the tree-level angle of the couplings of the mass fit, $\sin^2\theta_W = 0.22363$, changes the total by +0.82 %).

## 11 The dark sector

**Dark matter (a candidate whose relic abundance is OPEN).** The lightest odd states of $\Psi_A$ and of $\Psi_B$ (the same orbital in all eight blocks), two degenerate Dirac fermions of mass

$$M_{DM} = \varepsilon_{DM} = 3.10220\, m = 44.35 \text{ TeV} \qquad (m = H = 14.30 \text{ TeV}),$$

are cold ($w = 0$ at rest), electrically neutral and $Z$-neutral (UM-2 has no tip operator; a $P_+$ tip current would not reach $\Psi_B$ at all), and stable as long as the fold parity is exact (UM-2 has no oriented tip operator and the bulk is reflection symmetric) and $N_A$, $N_B$ are conserved (both hold in the good sector). Their hidden wave number carries 90 % of $\varepsilon^2$: their mass is the author's $x_0$ mechanism in the Heisenberg sense. Through I12 they

annihilate into $W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$ and gluon pairs. For $\mathcal{L} = G\,\bar\chi\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel (canonical fields) the spin-averaged cross-section is

$$\sigma v = \frac{G^2 s(s-4M^2)}{4\pi} \;\to\; \frac{G^2M^4v^2}{\pi} \quad (v \ll 1,\ v \text{ the relative velocity}), \qquad \text{in total} \quad \frac{M^4v^2}{\pi}\big(8G_g^2 + 3G_W^2 + G_B^2\big),$$

with $G_i = \mathcal{O}\,c_i\,g_i^2/\Lambda_\star^3$ ($g_i$ the four-dimensional gauge couplings, $\mathcal{O} = \partial\varepsilon_{DM}/\partial m = 0.7072$ the response of the dark-matter level to a uniform mass shift, DERIVED by the Hellmann-Feynman theorem). Since $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ does not mix the halves, each species freezes out separately. The standard freeze-out formulas (STANDARD: radiation domination, Dirac fermion with $g = 2$, $c = 1/2$, $\langle\sigma v\rangle = 6b/x$ with $x = M/T$ and $b = (M^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $g_* = 105.75$, $M_{Pl} = 1.22091\times10^{19}$ GeV)

$$x_f = \ln\Big[0.038\,c(c+2)\,g\,M_{Pl}M\langle\sigma v\rangle(x_f)\big/\sqrt{g_* x_f}\Big], \qquad \Omega h^2 = \frac{2\times1.07\times10^9\,\mathrm{GeV}^{-1}\,x_f}{\sqrt{g_*}\,M_{Pl}\,3b/x_f}$$

give $\Omega h^2 = 0.060$ per species, half of $\Omega_c h^2 = 0.120$, for $\langle\sigma v\rangle(x_f) = 2.01\times10^{-25}$ cm$^3$/s ($x_f = 30.38$), i.e. $(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (15.5\ \mathrm{TeV})^{-3}$ (a single species carrying all the dark matter would need $(17.6\ \mathrm{TeV})^{-3}$ with the same $g_*$). This requirement violates partial-wave unitarity (DERIVED): $\bar\chi\chi FF$ annihilates a fermion pair only from the $J = 0$, $P = +$ state ($^3P_0$), for which the spin-averaged inelastic cross-section of unpolarised Dirac fermions is bounded by

$$\sigma \le \frac{\pi}{4p^2} \quad (p \text{ the centre-of-mass momentum}), \qquad \sigma v \le \frac{\pi}{M^2 v}, \qquad \Big\langle\frac{\pi}{M^2v}\Big\rangle = \frac{\sqrt{\pi x}}{M^2} = 5.8\times10^{-26}\ \mathrm{cm}^3/\mathrm{s} \text{ at } x_f;$$

the requirement exceeds this bound 3.5 times, and the effective cross-section exceeds it for $v > 0.26$, i.e. for 79 % of the pairs at freeze-out. The gauge-kinetic portal therefore cannot produce the relic abundance by itself: an additional annihilation channel or a non-thermal origin of the dark matter is required (OPEN). Direct detection through I12 is far below current limits (even if the whole coupling were in gluons, $\sigma_{SI} \approx 6\times10^{-53}$ cm$^2$). Any relic value would also depend on the radion oscillation left over from the deflation (the moduli problem, section 14).

**Dark energy (NOT SUPPORTED).** No positive-energy state of $\Psi$ has an equation of state $w = p_3/\rho < 0$ on the static block; a $\lambda < 0$ condensate tuned to $w_0 = -1$ runs away ($w_a = -dw/da = -6$ at $a = 1$, the coefficient of the form $w = w_0 + w_a(1-a)$, $a$ the scale factor) and its energy density turns negative at redshift $z = 0.26$. The dark energy of UM-2 is the tuned residual $\Lambda_4$.

# 12 Cosmology: CMB, CNB, BAO

The four-dimensional effective theory is general relativity (section 8.5) with photons, the three neutrino mass eigenstates (1.000, 8.712, 50.140 meV, together $N_{eff} = 3.0432$), baryons, the dark matter (the odd quanta of $\Psi_A$ and $\Psi_B$) and $\Lambda_4$; there is no dark radiation. Its linear perturbations (order $\epsilon$) obey the standard Einstein-Boltzmann equations (STANDARD; Newtonian gauge, conformal time $\tau$, primes $d/d\tau$, $\mathcal{H} = a'/a$, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}]$, comoving wavenumber $k$):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2\delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar\rho + \bar P)\theta, \qquad k^2(\phi - \psi) = 12\pi G_N a^2(\bar\rho + \bar P)\sigma;$$

cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$; baryons and photons coupled by Thomson scattering ($n_e$ the free-electron density, $\sigma_T$ the Thomson cross-section, $R = 4\bar\rho_\gamma/(3\bar\rho_b)$, $c_s$ the baryon sound speed)

$$\delta_b' = -\theta_b + 3\phi', \qquad \theta_b' = -\mathcal{H}\theta_b + c_s^2k^2\delta_b + R\,an_e\sigma_T(\theta_\gamma - \theta_b) + k^2\psi,$$

$$\delta_\gamma' = -\tfrac{4}{3}\theta_\gamma + 4\phi', \qquad \theta_\gamma' = k^2\big(\tfrac14\delta_\gamma - \sigma_\gamma\big) + k^2\psi + an_e\sigma_T(\theta_b - \theta_\gamma),$$

$$F_{\gamma l}' = \frac{k}{2l+1}\big[lF_{\gamma(l-1)} - (l+1)F_{\gamma(l+1)}\big] - an_e\sigma_T\Big[F_{\gamma l} - \frac{\delta_{l2}}{10}\big(F_{\gamma2} + G_{\gamma0} + G_{\gamma2}\big)\Big] \quad (l \ge 2),$$

$$G_{\gamma l}' = \frac{k}{2l+1}\big[lG_{\gamma(l-1)} - (l+1)G_{\gamma(l+1)}\big] + an_e\sigma_T\Big[-G_{\gamma l} + \frac12\big(F_{\gamma2} + G_{\gamma0} + G_{\gamma2}\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big]$$

($F_{\gamma2} = 2\sigma_\gamma$; $G_{\gamma l}$ the polarisation multipoles); and one hierarchy per neutrino mass state $i$ (comoving momentum $q$, $E_i = \sqrt{q^2 + a^2m_i^2}$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed Fermi-Dirac distribution $\bar f_\nu$)

$$\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln\bar f_\nu}{d\ln q}, \qquad \mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}\big(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}\big) - \frac{E_ik}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q},$$

$$\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}\left[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}\right] \quad (l \geq 2).$$

They were solved with the Einstein-Boltzmann code CAMB, each neutrino mass state evolved separately with the degeneracy $N_{eff}/3$, at the Planck 2018 values (STANDARD) $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\,\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$, with $\theta$ held fixed (DERIVED): $H_0 = 67.37$ km/s/Mpc, $r_{drag} = 147.11$ Mpc, the TT and EE spectra agree with Planck ΛCDM (one neutrino of 0.06 eV, $N_{eff} = 3.044$) to $2.8 \times 10^{-4}$ ($30 \leq \ell \leq 2500$; for TT at most 0.007 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole). The BAO distances $D_M(z) = \int_0^z c\,dz'/H(z')$ (flat), $D_H = c/H(z)$ and $D_V = (zD_M^2 D_H)^{1/3}$, in units of the drag-epoch sound horizon $r_d$, are compared with the DESI DR2 measurements (STANDARD, arXiv:2503.14738)

| tracer | $z$ | $D_V/r_d$ | $D_M/r_d$ | $D_H/r_d$ | $r_{MH}$ |
|---|---|---|---|---|---|
| BGS | 0.295 | $7.942 \pm 0.075$ | | | |
| LRG1 | 0.510 | | $13.588 \pm 0.167$ | $21.863 \pm 0.425$ | $-0.459$ |
| LRG2 | 0.706 | | $17.351 \pm 0.177$ | $19.455 \pm 0.330$ | $-0.404$ |
| LRG3+ELG1 | 0.934 | | $21.576 \pm 0.152$ | $17.641 \pm 0.193$ | $-0.416$ |
| ELG2 | 1.321 | | $27.601 \pm 0.318$ | $14.176 \pm 0.221$ | $-0.434$ |
| QSO | 1.484 | | $30.512 \pm 0.760$ | $12.817 \pm 0.516$ | $-0.500$ |
| Ly$\alpha$ | 2.330 | | $38.988 \pm 0.531$ | $8.632 \pm 0.101$ | $-0.431$ |

through $\chi^2 = \sum_i (d_i - d_i^{th})^T C_i^{-1} (d_i - d_i^{th})$ with $C_i = \begin{pmatrix} \sigma_M^2 & r_{MH}\sigma_M\sigma_H \\ r_{MH}\sigma_M\sigma_H & \sigma_H^2 \end{pmatrix}$ for the pairs $d_i = (D_M/r_d, D_H/r_d)$ of the anisotropic tracers and $(d - d^{th})^2/\sigma_V^2$ for BGS: $\chi^2 = 31.7$ for 13 measurements (Planck ΛCDM: 32.6) (DERIVED).

# 13 Parameters

| parameter | value | status |
|---|---|---|
| $H$ | 14.30 TeV | DERIVED from $m_{KK}$, $\eta$ and $f_*$ |
| Lovelock weights $w$, $\Lambda$ | $(1, 1/32, 1/1152)$, $-10H^2$ | AUTHOR |
| $\eta = HL$ | 3 | ASSUMED (record) |
| $m = 3HM$ | $M = 1/3$ ($m = H$) | ASSUMED (record) |
| stabiliser law | $\rho_b \propto f^{-8}$, $p_0 = p_t = \rho_b = -p_3$ | ASSUMED |
| $k_*$, $f_*$, $G(k_*)$ | 21.522, 0.21556, 13.808 | DERIVED |
| $m_r$ | 48.4 TeV | DERIVED |
| $m_{KK}$ | 121.3 TeV | DERIVED (fit to $\rho_0 - 1$ and $m_W$) |
| $\beta_W$, $\beta_Z$; $\cos^2\theta_W$ | 0.011912, 0.015343; 0.776368 | DERIVED (fit; $\beta_Z = \beta_W/\cos^2\theta_W$, $\theta_W$ the angle of the couplings, fixed by $m_W/m_Z$ and $\rho_0$) |
| tip matrices $\kappa_e$, $\kappa_u$; $\kappa_\nu$; $\kappa_d$ | 1; $U_{PMNS}$; $V_{CKM}$ (unitary) | ASSUMED (the mixing values STANDARD, measured) |
| doublet localisations $c_L$, $c_Q$ | 5.9529, 3.3823 | DERIVED (fit to $m_\tau$, $m_t$ with $c_R = -c_L$) |
| singlet localisations $c_R$ | $\nu_{1,2,3}$: $-35.045$, $-32.848$, $-31.070$; $e$, $\mu$, $\tau$: $-14.554$, $-8.981$, $-5.953$; $u$, $c$, $t$: $-15.430$, $-8.768$, $-3.382$; $d$, $s$, $b$: $-14.626$, $-11.516$, $-7.501$ | DERIVED (fit to the masses) |
| lightest neutrino mass $m_1$; ordering | 1.000 meV; normal | ASSUMED |

| parameter | value | status |
|---|---|---|
| $\Delta m^2_{21}$, $\Delta m^2_{31}$; $\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, $\delta$ | $7.49\times10^{-5}$, $2.513\times10^{-3}$ eV$^2$; 0.308, 0.470, 0.02215, 212° | STANDARD (NuFIT 6.0) |
| $N_{eff}$ | 3.0432 | STANDARD (precision value; UM-2 computes 3.0476 to 3.0485) |
| $c_Z$, $c'_Z$ (the tip $Z$ portal of version 1) | 0 (removed) | ASSUMED (bounded by $N_\nu$, disfavoured by $N_{eff}$) |
| $c_V$ (I11) | 0 (benchmark) | ASSUMED |
| I12 coefficients | $(8G_g^2+3G_W^2+G_B^2)^{1/2}=(15.5\text{ TeV})^{-3}$ would be the freeze-out requirement (two species) | DERIVED; it violates the $^3P_0$ unitarity bound, so the relic is OPEN |
| $M_{DM}$ | 44.35 TeV | DERIVED |

# 14 Assumptions, open problems and conflicts with data

- **ASSUMED**: the regular tip cutoff $\eta=3$; the good sector and compact extra times; $m=H$; the block stabiliser and its $f^{-8}$ law; the boundary assignments of the Standard-Model fields; the tip sigma field; one doublet bulk mass per sector, common to the three generations, with $c_R=-c_L$ for $\tau$ and $t$; unitary tip matrices $\kappa_e=\kappa_u=1$, $\kappa_\nu=U_{PMNS}$, $\kappa_d=V_{CKM}$; the normal ordering with $m_1=1$ meV; the solar density profile and the supernova envelope profile of the flavour computations; the record's particle convention for the Krein metric.
- **INVENTED**: the portals I11 and I12 (the chiral tip $Z$ portal I9 of version 1 is removed).
- **OPEN**: the microscopic origin of the stabiliser; the tachyonic extra-time modes outside the good sector; the closed time-like curves implied by compact extra times; the radion oscillation left over from the deflation (ESTIMATE: its Planck-suppressed decay rate $\Gamma=C_r\,m_r^3/(4\pi\bar{M}_{Pl}^2)$, with $\bar{M}_{Pl}=2.435\times10^{18}$ GeV and an order-one coupling factor $C_r$, gives a lifetime of $0.43\,\text{s}/C_r$ and the reheating temperature $T_{RH}=(90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma\bar{M}_{Pl}}=1.85\sqrt{C_r}$ MeV ($g_*=10.75$), below the 4.7 MeV that nucleosynthesis needs unless $C_r>6.5$: the cosmological moduli problem, which also conditions the dark-matter relic); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the $S$ parameter of the Higgsless tip; anomalies of the eight-dimensional bulk theory; the cosmological constant ($\Lambda_4$ tuned to $10^{-92}H^2$); the electroweak sector beyond tree level (the angle of the couplings of the $W^\pm$, $Z$ mass fit, $\sin^2\theta_W=0.22363$, differs from the $\overline{MS}$ value 0.23122 used in the $Z$ couplings by the Standard-Model loop corrections, which UM-2 does not compute; the total tree-level $Z$ width changes by 0.82 % between the two, and the computed total, 2485.2 MeV, lies 0.4 % below the measured value); the absolute neutrino mass scale and the mass ordering (only bounded, and preferred, by data); the collective flavour evolution of supernova neutrinos beyond the linear analysis; the dark-matter relic abundance (the freeze-out requirement for I12 exceeds the $^3P_0$ unitarity bound 3.5 times: an additional annihilation channel or a non-thermal origin is needed).
- **CONFLICTS WITH DATA, inherent in the order**: there is no Higgs boson while one is observed at 125 GeV; without it the longitudinal $W^+W^-$ scattering violates perturbative unitarity near $\sqrt{8\pi}\,v\approx1.2$ TeV, far below the Kaluza-Klein modes that could restore it. (The one-generation conflict of version 1, a total $Z$ width of 1.23 GeV, is removed by the three generations.)